

1 Introduction

One of the central achievements of the Minimalist Program lies in the way that it has reformulated the notion of DISPLACEMENT. Breaking from the classical stance of GB, Chomsky 1993 proposes that movement in the syntax does not substitute the elements that it targets for *traces*. This theoretical step emerges directly from two guiding principles of minimalist inquiry: (i) the evolutionary mandate to minimize the number of operations and objects available to the syntax, and (ii) the conceptual hypothesis that syntactic operations do not affect the elements that they target, for the reasons of economy expressed by the INCLUSIVENESS CONDITION (Chomsky, 1995) and the NO-TAMPERING CONDITION (Chomsky, 2008). The ensuing theoretical gain, in turn, has since been shown to come hand-in-glove with a wide range of empirical advances: first in the domain of reconstruction (Chomsky, 1993) and later at the interface of syntax and phonology. In this second arena, specifically, this move has opened up a rich and impressive series of advances into two types of interface phenomena: the realization of moved elements in multiple positions along their paths (Landau, 2006; Kandybowicz, 2007; Bleaman, 2022) and the realization of moved elements in positions determined by the phonology (Bobaljik, 1995, 2002; Bošković, 2001, 2002).

The goal of this paper is to leverage this theoretical understanding to document and analyze an apparent interaction between phrasal movement and the phrasal phonology. This is one that emerges in the clausal syntax of Mandar, an Austronesian language of the South Sulawesi subfamily which shows an otherwise-consistent basic word order of vso. In the clauses of this language that contain preverbal auxiliaries, the absolutive argument canonically appears in its ordinary postverbal position (1a). But if we increase the number of word-level stresses that fall on other constituents in the VP—for instance, by reduplicating the verb—the absolutive argument seems to move to a position above the verb (1b).

(1) *Movement and Stress in Mandar*

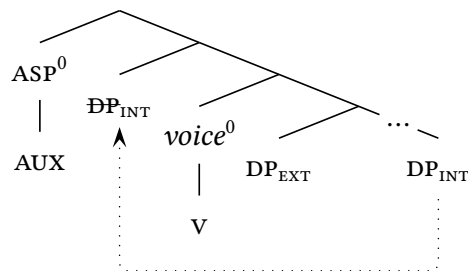
- a. Rua i [VP na-waca *pro*_{ERG} **Barriers**_{ABS}].
 once 3ABS 3ERG-read he BOOK
 ‘Once he read Barriers.’
-
- b. Rua i **Barriers**_{ABS} [VP nawa-na-waca *pro*_{ERG} ____].
 once 3ABS BOOK RED-3ERG-read he
 ‘Once he sorta read Barriers.’

The result is a system that, at first blush, seems to run against the principle of PHONOLOGY-FREE SYNTAX (Zwicky & Pullum, 1986): in Mandar, a step of movement seems to target

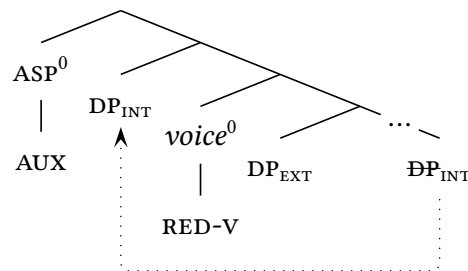
the absolutive argument in response to phonological changes in other elements in the VP. But as our investigation unfolds, we will see that this effect must be understood as a case where phonological factors determine where an element is spelled out along its path of movement in the syntax. In Mandarin, I will propose that the absolutive argument systematically undergoes the step of movement in (2a): one which places it at the edge of the extended VP, on a par with similar processes in other languages where the absolutive argument receives Case from τ^0 (Aldridge, 2004; Coon *et al.*, 2014). In the default case, I will argue, this movement is usually *covert*: the absolutive argument is typically realized in its base position, yielding the default vso word order of the language. But as the prosodic weight of the material in the VP is increased—and as the VP begins to fall under prosodic requirements of BALANCE—the absolutive argument must be spelled out in its higher position on this path (2b).

(2) *Phonologically-Conditioned Variation in Spell-Out*

a. *Default Case: Low Spell-Out*



b. *Marked Case: High Spell-Out*



The result is an analysis, in the tradition of Franks & King 2000 and Bošković 2001, which recognizes that output constraints must be able to guide the reduction of movement dependencies. Building from the theory of the syntax-prosody interface of Selkirk 2009, I develop a theory of how these dependencies are linearized in the phonological calculus where prosodic structure is built: one which draws from the theory of MULTIDOMINANCE (Kracht, 2001) to reduce these dependencies through a combination of the regular constraints of the phonology, the pressure to realize moved elements in the highest possible positions (Brody, 1995), and the interface requirements of particular heads (Landau, 2007).

The remainder of this paper is structured as follows. Section 2 introduces Mandarin and lays out the essential syntax of the clause and introduces the interaction of interest; Section 3 argues that it reflects spell-out of the absolutive DP in a position it always moves to in the syntax. Section 4 shows how this pattern has a transparently output-optimizing prosodic effect. Section 5 lays out the theory of this interaction; Section 5 concludes.

2 Background

Mandar is an Austronesian language spoken by 400,000 people in the Indonesian province of West Sulawesi. It belongs to the South Sulawesi subfamily of Malayo-Polynesian (Esser, 1938; Mills, 1975; Smith, 2017) and can be productively compared to the other languages of that subfamily, which have been described in a larger body of English-language work (Friberg, 1991, 1996; Strømme, 1994; Matti, 1994; Valkama, 1995a,b; Jukes, 2006; Lee, 2008; Kaufman, 2008; Laskowske, 2016; Finer, 1994, 1997, 1998, 1999; Béjar, 1999; Finer & Basri, 2020). Mandar has been the subject of an Indonesian-language grammar (Pelenkahu *et al.*, 1983), a survey of adverbs (Sikki *et al.*, 1987), a compilation of traditional poetry (Muthalib & Sangi, 1991), and a conversational handbook (Friberg & Jerniati, 2000), as well as an English-language survey of dialects (Grimes & Grimes, 1987). This paper will focus on the standard variety of the language, which is spoken on the coast between the cities of Polewali and Majene (Grimes & Grimes, 1987). Its results emerge from long-term research with a single native speaker, Jupri Talib, and systematic cross-checking with two native speaker linguists: Sitti Sapiah (University of West Sulawesi) and Hairuddin (Universitas Hasanuddin). The language of contact in all of this primary work has been Indonesian.¹

Mandar shows a consistent unmarked word order of v-s-o-d-x (Brodkin, To Appear) (verb > external argument > internal argument > applied argument > vp-level adjuncts). The language shows no morphological case, but it does have an ergative-absolutive system of agreement: in clauses that contain transitive verbs, the verb hosts a prefix that spells out agreement with the external argument (EXT) and an absolutive enclitic spells out agreement with the internal argument (INT). Example (3) shows a clause of this shape.

(3) *The Mandar Clause*

V		S	O		
U-waca	i	yau	karewa	angga'-u	monge'.
1ERG-read	3ABS	1SG	news	until-1ABS.NONFIN	sick

‘I read the news until I got sick.’

The Mandar verb also shows alternations in *voice* that determine the identity of the argument that is targeted by absolutive agreement. I will underline this argument all ex-

¹GLOSSING: ABS: absolutive, ANTIP: ACC: accusative, APPL: applicative, DAT: dative, ERG: ergative, GEN: genitive; 1SG: first-person singular. ORTHOGRAPHY: <c, ng, '>: /tʃ, ŋ, ?/.

amples below. For instance, transitive verbs can be antipassivized by replacing the ergative agreement with the prefix *man*. In finite clauses that contain antipassive verbs, like the antipassive version of example (3) below, absolutive agreement must target the EXT.

(4) *The Voice Alternation in Mandar*

V	S	O		
Mam -baca a'	<u>yau</u>	karewa	angga'-u	monge.
ANTIP-read 1ABS	1SG	news	until-1ABS.NONFIN	sick
'I read news until I got sick.'				

This starting alternation between transitive and antipassive verbs is mirrored by a much wider network of alternations in voice in Mandar that collectively constitute what is known as a Western Austronesian Voice system: a series of morphological alternations on the verb, bound up with the *voice*P-level syntax of transitivity, that influence the clause-level pattern of abstract Case assignment (and many more systems that turn on that) (Guilfoyle *et al.*, 1992; Aldridge, 2004; Nomoto, 2013, 2015). The voices that make up this system in Mandar have been described in a growing body of work (Brodkin 2021a, 2022b), and in the present investigation, it will be useful to become familiar with exactly one more. This is the ditransitive voice, which is morphologically built from the circumfixion around the verb of the ergative agreement prefixes in (3) and the applicative suffix in example (5). In this voice, the pattern of absolutive agreement must target the applied argument (GOAL).

(5) *The Ditransitive Voice*

	V	S	O	D
Apa na	mu -waca- ngang a'	i'o	karewa	<u>yau</u> ?
What and	2ERG-read-APPL 1ABS	2SG	news	1SG
'Why are you reading me the news?'				

2.1 The System of Auxiliaries

The work of our paper begins with the introduction of a closed class of preverbal elements that I will refer to as auxiliaries. These are rigidly-ordered elements that mark distinctions of polarity and aspect, and there has been a steady accumulation of evidence to suggest that they form a coherent class in the syntax (Brodkin 2021a,b, 2022a,b). Several elements of this set are presented in fragment contexts in the following set of examples: the realis clausal negator *ndang* (6a), the experiential perfect *rua* (6b), and the completive *yari* (6c).

(6) *Three Mandar Auxiliaries*

a. Ndang na-waca	b. Rua na-waca	c. Yari na-baca
not 3ERG-read	once 3ERG-read	DID 3ERG-read
‘Not reading’	‘Once read’	‘Did read’

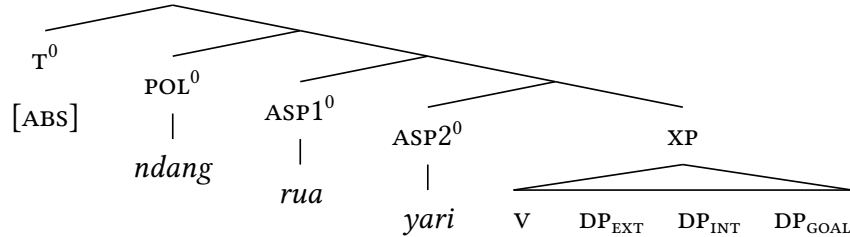
The auxiliaries are initially interesting for the way that they interact with absolutive agreement. In the finite clauses that contain auxiliaries, it is ungrammatical for absolutive agreement to follow the verb. Instead, it must surface after the highest overt auxiliary: in example (7), the negator *ndang* (and for discussion of the identical pattern across the subfamily, see [Strømme 1994](#); [Matti 1994](#); [Valkama 1995a](#); [Friberg 1996](#); [Finer 1999](#)).

(7) *Absolutive Agreement: Follows the Highest Aux*

Ndang i	rua	yari	mam-baca.
not 3ABS	once	DID	ANTIP-read
‘She never DID read.’			

The natural way to interpret this body of facts, after especially [Brodkin 2022a](#), is this: (i) the auxiliaries spell out a set of functional heads in the middle field (POL > ASP > ASP2), (ii) absolutive agreement originates in a head that sits above them in the syntax (T), and (iii) the absolutive agreement enclitic surfaces after the highest overt head in that space: the highest auxiliary whenever auxiliaries are present and the verb whenever they are not. Setting aside the exact mechanisms of its linearization, the result is the syntax in (8).

(8) *Mandar Clause Structure*



We can now leverage this sketch of the middle field to pin down the exact syntax of the vsOD string. Implicit in tree (8) is the claim that there is a single constituent—here labeled XP—that sits beneath the auxiliaries and contains the verb and all arguments that follow. If this analysis is correct, then we should expect that strings of the order vsOD will be able to pass constituency tests in the syntax. And as a matter of fact, they can. To open the case, we can note that many types of adjuncts—like locative and temporal expressions—canonically fall at the right edge of the vsOD string. This fact is shown in example (9).

(9) *The VSO string: Final Adjuncts*

Yari a'	[_{XP} na-waca-ngang	iKaco'	kalinda'da'	]	dionging.
DID 1ABS	3ERG-read-APPL	NAME	Mandar poem		yesterday

'Kaco' DID read me a poem yesterday.'

If the vsOD string does indeed form a constituent, then it seems natural to imagine that adjuncts like *dionging* ‘yesterday’ will be right-adjoined to that xP in clauses like (9). On that view, we can make a prediction about ellipsis: if there is any operation like VP-ellipsis in the language, then it might be able to elide the vsOD string while sparing the auxiliaries that precede it and the adjuncts that follow. The following example shows that this prediction is correct: there is indeed a process of VP-ellipsis that operates under the auxiliaries in Mandar, it targets the vsOD string, it is able to spare the adjuncts that follow.

(10) *The VSO string: Ellipsis*

- a. Makany_{NP}nyang a' mua' melo a' na-waca-ngang iKaco' kalinda'da'.
 sure 1ABS that want 1ABS 3ERG-read-APPL NAME Mandar poem
 'I was sure that Kaco' wanted to read me a poem.'
- b. Na yari a' [_{XP} [_{XP} ~~na-waca-ngang~~ ia mesa] dionging].
 and DID 1ABS 3ERG-read-APPL he one yesterday
 'And he DID ___ yesterday.'

In a close study of this system (and the parallel syntax of predicate clefting), Brodgin To Appear thus makes the following case: when we control for the occurrence of scrambling (which proceeds to the right and places arguments after adverbs like *dionging* “yesterday”), Mandar strings of the order vsOD strictly forms tight constituents in the syntax. On the basis of finer patterns of prosodic phrasing, moreover, Brodgin To Appear demonstrates that this xp is in fact made up of a sequence of smaller and nested constituents. The largest of these is a constituent that contains the substring sOD. This constituent, in turn, contains a smaller projection that surrounds the substring oD. On the basis of these facts, Brodgin To Appear argues that the vsOD string shows the rough internal syntax in (11).

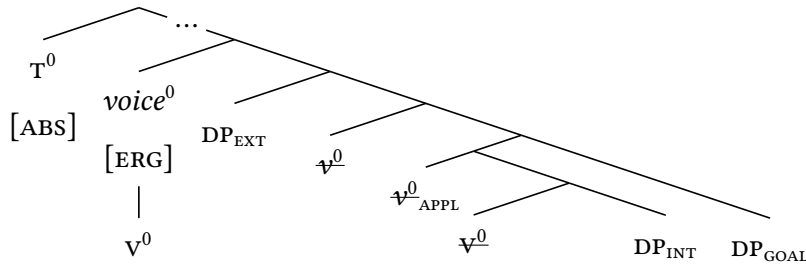
(11) *The Structure of the VSOD String*

$$[{}_{\text{TP}} \text{ ABS AGR } [{}_{\text{POLP}} \text{ NEG } [{}_{\text{ASP1}} \text{ AUXES } [{}_{\text{ASP2}} \text{ AUXES } [{}_{\text{XP}} \text{ V } [{}_{\text{YP}} \text{ S } [{}_{\text{ZP}} \text{ O D }]]]]]]]]$$

We can work this initial sketch into a theory of clause structure with a short turn to the system of verbal morphology. [Brodkin 2022b](#) and [Brodkin 2025](#) demonstrate that the

Mandar verb carries a string of affixes that are distributed across at least three distinct heads: v^0 (which I take to introduce the EXT; Collins 2005; Merchant 2013), $appl^0$ (which introduces the GOAL; Pykkänen 2008), and $voice^0$ (which hosts ergative agreement in the transitive and ditransitive voices and a subpart of the affix *man*- in the antipassive). Taking the verb to pick up this morphology via head-movement, we arrive at the clausal syntax in (12): in a vsOD clause that contains a ditransitive verb, the verb raises to $voice^0$ and the postverbal arguments are all spelled out in their base positions within the vp .

(12) *Mandar Syntax: the Ditransitive*



We can now conclude this introduction by acknowledging a process of rightward displacement that we will control for and set aside in the discussion below. When postverbal arguments appear before *voiceP*-level adjuncts in Mandarin, they must take the order vsOD (13a). But it is possible for definite arguments to surface in any order in positions that follow *voiceP*-level adjuncts, and in this second context, Brodtkin 2025 shows that they have been scrambled to the right (13b). To control for scrambling and ensure that strings of the order vsOD will always form constituents, then, we will restrict our attention to clauses where all postverbal arguments fall before *voiceP*-level adjuncts of this type.

(13) *Scrambling to the Right*

- a. Na-alli-ang i [s iKaco'] [o iGary] [d iCicci'] dio.
 3ERG-buy-APPL 3ABS NAME NAME NAME there
 ONLY vsODX: 'Kacho' bought Gary (a cat) for Chichi' there.'
- b. Na-alli-ang i dio [?? iKaco'] [?? iGary] [?? iCicci'].
 3ERG-buy-APPL 3ABS there NAME NAME NAME
 Possible interpretations: vxSOD, vxSDO, vxOSD, vxDSO, vxODS, vxDOS

We then proceed into our core empirical terrain with these foundations in place:

- (i) The post-auxiliary vsOD(x) string forms a constituent that corresponds to the *voiceP*,
- (ii) the auxiliaries spell out a string of functional heads that sit just above the *voiceP*,
- (iii) the head that triggers absolutive agreement sits immediately above all of these heads.

2.2 A Step of Movement

Our phenomenon of interest emerges in an interaction between linear order and stress. In an idiosyncratic quirk of phonology, the Mandar antipassive prefix shows an alternation that impacts the distribution of word-level stress within the verbal complex. The default form of the antipassive prefix is *maŋ-*, and this is the form that we have seen in all of the examples up to this point. But whenever this prefix surfaces before roots that begin with the segment /a/, it reduces to the subsyllabic form *m-*. This change, in turn, has an impact on the distribution of stress: when the antipassive prefix is realized as *maŋ-*, it carries a separate instance of word-level stress (14a), and when it is realized as *m-*, it does not (14b).

(14) *The Antipassive Prefix: Allomorphy and Stress*

- a. **mám**-báca, **mán**-dúndu, **máng**-gáli
 ANTIP-read ANTIP-drink ANTIP-dig
 ‘read, drink, dig’
- b. **m**-ánde, **m**-ála, **m**-allái
 ANTIP-eat ANTIP-take ANTIP-scold
 ‘eat, take, scold.’

This alternation has a regular phonological shape, and so it seems natural to situate it—after *Brodkin To Appear*—in the regular phonology of the language. But in clauses that contain auxiliaries, it provokes an immediate change in the order of constituents. In the clauses that contain an auxiliary, an EXT, and an antipassive verb that begins in /a/, the only grammatical word order is the ordinary and expected AVS (15a). But in the clauses that contain an auxiliary, an EXT, and an antipassive verb that begins in any other segment, the unmarked word order becomes ASV (15b): the EXT must raise above the verb.

(15) *The Prefixal Alternation: A Change in Word Order*

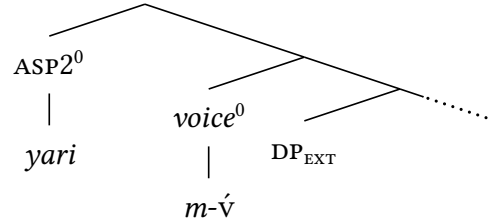
- a. Yári i [voiceP **m**-ánde iÁli].
 DID 3ABS ANTIP-eat NAME
 ‘Ali DID eat.’
- b. Yári i [voiceP iÁli **mán**-dúndu ____].
 DID 3ABS NAME ANTIP-drink
 ‘Ali DID drink.’

The force of this pattern is to suggest that the grammar of Mandar contains a process of the following shape: in antipassive clauses that contain auxiliaries, the EXT must raise to the edge of the *voiceP* if the verbal complex contains two instances of word-level stress.

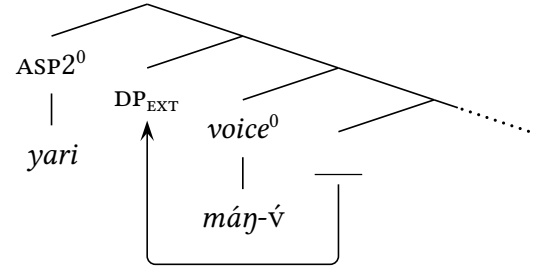
We can schematize this interaction along the following lines.

(16) *The Step of Movement*

a. *Single Stress: EXT = Low*



b. *Two Stresses: EXT = High*



We thus arrive at a generalization that raises immediate and serious questions about the architecture of the grammar: a step of phrasal movement seems to be conditioned by the distribution of word-level stress. To work through the meaning of this state of affairs, then, it will be useful to establish two further generalizations about its immediate shape.

The first generalization surrounds the arguments that undergo this step of movement. In the examples in (15), we have seen that the introduction of the stressed antipassive prefix *man-* triggers movement of the EXT. In this first case, this step of movement targets an EXT that is syntactically complex: the DP *iAli* consists of a name and the proprial article *i*. But this process applies in the same way to EXTs of all different syntactic shapes: for instance, bare nominals that host the null definite determiner (17a) and DPs that contain overt demonstratives (17b) (on the exact syntax of these elements: Brodtkin 2022b, 2025).

(17) *The Movement Rule: Targets All Types of EXT*

- a. Yári i [voiceP gúru mán-dúndu ____].
 DID 3ABS teacher ANTIP-drink
 ‘The teacher DID drink.’
- b. Yári i [voiceP iting gúru mán-dúndu ____].
 DID 3ABS that teacher ANTIP-drink
 ‘That teacher DID drink.’

There is only a single restriction that blocks the movement of the EXT in this context, and it is one of the phonology. In the examples above, each EXT carries a single instance of word-level stress—falling on *Ali* in (15b) and on *guru* in (17a)-(17b). This is a strict prerequisite for the step of movement in (16), and in every case where the EXT has a different shape, it is forced to remain in the *voiceP*. The following examples illustrate.

There are a number of DP-internal elements that carry independent word-level stress in Mandarin, including the reduplicant (18a) and all adjectives (18b). When these elements are added to the EXTs in (17), our pattern of movement is suspended—and DPs like *gúru-gúru* ‘the teachers’ and *iting gúru malinggáo* ‘that tall teacher’ are forced to follow the verb.

(18) *The Movement Rule: Suspended when the EXT is Heavy*

- a. Yári i [voiceP ____ mán-dúndu gúru-gúru].
 DID 3ABS ANTIP-drink RED-teacher
 ‘The teachers DID drink.’
- b. Yári i [voiceP ____ mán-dúndu iting gúru malinggáo].
 DID 3ABS ANTIP-drink that teacher tall
 ‘That tall teacher DID drink.’

Setting aside the hypothetical scenario in which the EXT carries NO word-level stress (which is unattested in Mandar), we thus arrive at the descriptive generalization in (19).

(19) *The First Generalization*

In clauses that contain auxiliaries and antipassive verbs that carry the stressed prefix *mán-*, the EXT raises above the verb, no matter its internal syntactic shape, IF AND ONLY IF it carries a single instance of word-level stress.

We can now lay out a second generalization on the shape of the trigger for this effect. In the cases that we have studied so far, the EXT has been forced to raise by the introduction of a second stress within the verbal complex. But the same process of movement can also be forced by introducing a second instance of word-level stress anywhere within the *voiceP*. For instance, in the clauses that contain auxiliaries, it is possible for antipassive verbs of all shapes to be followed by an INT (20a). Slightly foreshadowing the discussion to come, we can note that it is ungrammatical here for our process to target the INT (20b).

(20) *Antipassive Verbs + Internal Arguments*

- a. Yári i [voiceP mán-dúndu bálló'].
DID 3ABS ANTIP-drink palm wine
'She DID drink palm wine.'
- b. *Yári i [voiceP **bálló'** mán-dúndu ____].
DID 3ABS palm wine ANTIP-drink
INTENDED: 'She DID drink palm wine.'

When an EXT is introduced into a clause that contains an auxiliary, an antipassive verb, and an INT that carries one word-level stress, it is ungrammatical for the EXT to surface in its usual position between the v and the INT. In this context, rather—even when the verb carries a single instance of word-level stress—the EXT is forced to raise (21).

(21) *Antipassive Verb + Internal Argument: Forces Movement of the EXT*

Yári i [voiceP iMína m-ánde ____ ánde].
 DID 3ABS NAME ANTIP-eat rice
 ‘Mina DID eat rice.’

The introduction of *voiceP*-external adjuncts, however, does not trigger the same effect: when an EXT is introduced into a clause that contains an auxiliary, an antipassive verb, and an x that carries one word-level stress, the EXT is forced to remain in its base position.

(22) *Antipassive Verb + VoiceP-Level Adjunct: XMovement of the EXT*

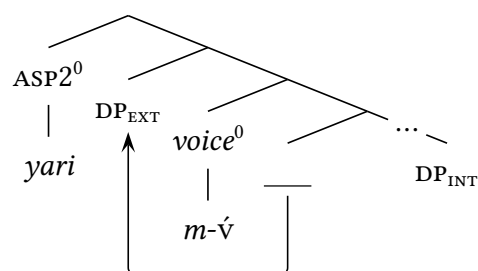
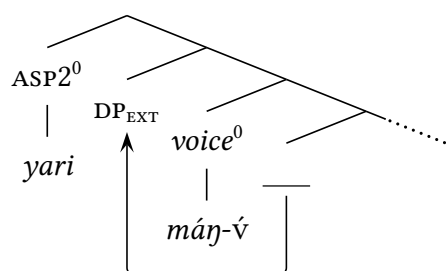
Yári i [voiceP ____ m-ánde iMína] diónging.
 DID 3ABS ANTIP-eat NAME yesterday
 ‘Mina DID eat yesterday.’

We thus seem to be facing down the situation in (23): in clauses that host auxiliaries, the EXT moves whenever it carries one instance of stress and the remaining material in the *voiceP* carries two—either in the verbal complex (23a) or between the v and INT (23b).

(23) *The Step of Movement: Conditioned by Stress within the VoiceP*

a. *Two Stresses in the Verbal Complex*

b. *Two Stresses between the V + INT*



On the understanding that we will continue to refine this descriptive picture as our investigation proceeds, we can summarize the revised shape of our pattern as (24).

(24) *The Second Generalization*

In clauses that host auxiliaries and antipassive vs, the EXT is forced to raise when
 (i) the EXT carries a single instance of word-level stress and
 (ii) the *voiceP* would otherwise contain three instances of word-level stress.

2.3 Absolute Preposing

The descriptive reformulation in (24) suggests that we are facing down a type of movement in the *voiceP* that is directly conditioned by the distribution of word-level stress. But it seems very doubtful that any operation of the syntax could be genuinely motivated by phonological factors in this way: almost all known syntactic operations are insensitive to the phonology (the principle of PHONOLOGY-FREE SYNTAX; Zwicky & Pullum 1986), and there is good amount of evidence to suggest that phonological information is not even present until a very late derivational stage—at the absolute earliest, at the tail end of the postsyntactic morphology (Halle & Marantz, 1993; Arregi & Nevins, 2012). It thus seems natural to ask whether our effect might actually reflect a matter of the phonology—and more specifically, whether it might be recast as the type of long-distance phrasal movement been argued to occur in the phonology by Agbayani & Golston 2010, 2016, Agbayani *et al.* 2015, Clemens 2014, 2019 and Clemens & Coon 2018.

There is reason to believe, however, that this pattern of movement is not a purely phonological effect. This is because its shape is equally conditioned by the systems of the syntax—and more particularly, by the clause-level pattern of absolute agreement. To see this fact, it will be useful to turn to clauses that host transitive verbs. The following example presents a clause of this shape: there, the transitive verb *nawaca* “read” contains a single instance of word-level stress and is followed by the INT *buku* “the book.” The transitive INT triggers absolute agreement and the word order here must be AUX-V-DP.

(25) *Transitive + Ditransitive Verbs: Single Instance of Stress*

Rúa i	[<i>voiceP</i> na-wáca	<u>búku</u>].
once 3ABS	3ERG-read	book

‘He once read the book.’

Once we introduce an additional word-level stress into the *voiceP*—for instance, by adding an EXT—the language once again begins to prohibit the order AVSO (26).

(26) *Transitive Clauses: Ungrammatical Outcomes*

*Rúa i	[<i>voiceP</i> na-wáca	iÁli	<u>búku</u>].
once 3ABS	3ERG-read	NAME	book

INTENDED: ‘Ali once read the book.’

The repair in this context, however, is not to move the EXT (27a): when the verb is transitive, rather, this prosodic configuration seems to force movement of the INT (27b).

(27) *Transitive Clauses: Movement of the INT*

- a. *Rúa i [voiceP iÁli na-wáca — búku].
 once 3ABS NAME 3ERG-read book
 INTENDED: ‘Ali once read the book.’
- b. Rúa i [voiceP búku na-wáca iÁli —].
 once 3ABS book 3ERG-read NAME
 ‘Ali once read the book.’

Just like the process that raises the antipassive EXT, this step is suspended when the transitive INT contains more than one stress—for instance, when it hosts an adjective. And just like the process that raises the antipassive EXT, this step places its targets exactly between the verb and all higher auxiliaries. In the clauses that contain multiple auxiliaries, the transitive INT and the antipassive EXT must each be placed beneath the lowest AUX.

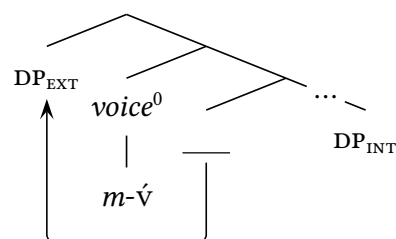
(28) *A Shared Landing Site: Beneath the Lowest Aux*

- a. Rúa i yari [voiceP iÁli m-ála — kursus].
 once 3ABS DID NAME ANTIP-take course
 ‘Ali once DID take a course.’
- b. Rúa i yari [voiceP kúrsus na-ála iÁli —].
 once 3ABS DID course 3ERG-take NAME
 ‘Ali once DID take the course.’

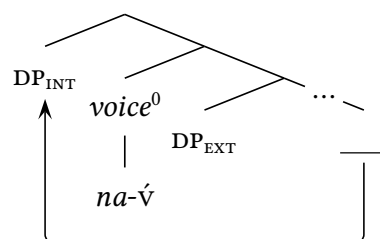
The result is that we seem to be dealing with a mirror image of the step in Section 2.2: in the contexts where the EXT is forced to raise in the clauses whose verbs are antipassive, the INT is forced to raise in clauses whose verbs are transitive.

(29) *Parallel Steps of Movement: The Antipassive and Transitive Voices*

a. *Antipassive: Raising of the EXT*



b. *Transitive: Raising of the INT*



We can round out this investigation with a turn to the ditransitive, where absolutive agreement targets the GOAL. In clauses of this type, the EXT and the INT always stay low. In ditransitive clauses that host AUXES, EXTs, and INTs, then, the word order must be AVSO.

(30) *Ditransitive Clauses: Postverbal EXTs and INTs*

Rúa i [voiceP na-allí-ang iÁli lo'díang].
 once 3ABS 3ERG-buy-APPL NAME ring
 'Ali once bought her a ring.'

Under the usual prosodic conditions, however, ditransitive clauses ban the linear orders AVOD and AVSD—orders in which the *voiceP* hosts three instances of stress and one falls on the GOAL (31a). The repair is presented in (31b): our usual step targets the GOAL.

(31) *Ditransitive Clauses: Problems with Postverbal GOALS*

- a. *Rúa i [voiceP na-allí-ang lo'díang iCícci'].
 once 3ABS 3ERG-buy-APPL ring NAME
 INTENDED: 'He once bought Cicci' a ring.'
- b. Rúa i [voiceP iCícci' na-allí-ang lo'díang ____].
 once 3ABS NAME 3ERG-buy-APPL ring
 'He once bought Cicci' a ring.'

The result is a direct correlation between this systems of movement and absolutive agreement: this process targets all and only all the DPs that control absolutive agreement. The force of this correlation, in turn, is to suggest that the effects of this section are closely bound up with the particular syntax of the DP that controls absolutive agreement: only this DP is forced to surface above the verb when it carries one word-level stress and the remainder of the *voiceP* contains two more. Labeling this DP the *absolutive DP* and the ensuing process *Absolutive Preposing*, we thus revise our descriptive summary to (32).

(32) *Absolutive Preposing: Revised Definition*

- In the clauses that contain auxiliaries, there is a step of movement that targets the DP that controls absolutive agreement (henceforth the absolutive DP) when:
- (i) the absolutive DP carries a single instance of word-level stress, and
 - (ii) the *voiceP* would otherwise contain three instances of word-level stress.

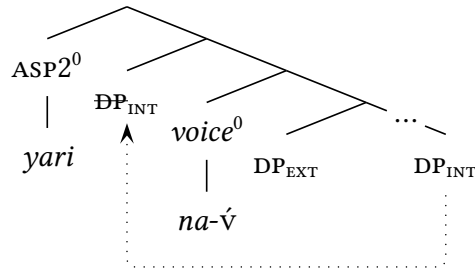
In light of this profile, I will assume that Absolutive Preposing must emerge from the joint work of two systems in the grammar of Mandar: the syntactic systems that position the absolutive DP and the phonological systems that govern the distribution of stress. It falls to the following sections, then, to understand how this interaction might play out.

3 The Syntax of the Absolutive DP

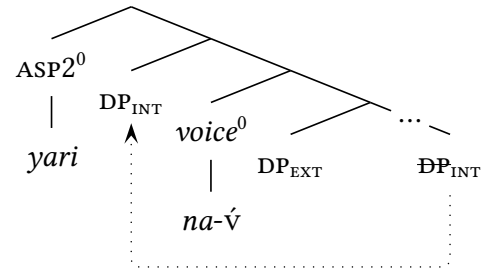
To capture these facts, I propose that there is a regular syntax beneath this system: in Mandar, the absolutive argument systematically raises to SPEC,voiceP . On this analysis, the descriptive effect of Absolutive Preposing must then reduce to a surface-oriented fact of spell-out: the absolutive DP is typically spelled out in its base position, yielding orders like AVO in transitive clauses (33a), but under the right prosodic conditions, it is spelled out in SPEC,voiceP instead—yielding preposing orders like AOVs (33b).

(33) *Absolutive Preposing = Variation in Spell-Out*

a. *Default Case: Low Spell-Out*



b. *Marked Case: High Spell-Out*



3.1 The Case for Covert Movement

The case for this step of movement in the syntax begins with pronominal coreference. Brodtkin 2022b shows that the grammar of Mandar contains a principle like Condition C (Chomsky, 1981; Reinhart, 1983): pronouns cannot c-command coreferential R-expressions. Thus pronominal EXTs, for instance, cannot be coindexed with R-expressions in lower PPs.

(34) *Mandar: Ordinary Condition C*

Sicaka i [EXT *pro*] [PP sola nyawa-na kanne'-na iAli].
 convene 3ABS he with soul-3GEN grandfather-3GEN NAME
 'He_{*i,j} convened with the ghost of Ali_i's grandfather.'

This restriction suggests the presence of a constraint with the shape in (35).

(35) *Condition C*

A pronoun cannot c-command a coreferential R-expression in Mandar.

This restriction is important for what it reveals about the syntax of the absolutive DP. In the contexts where neither the EXT nor the INT control absolutive agreement on τ^0 —for

instance, in the ditransitive—Mandar shows a restriction on coreference that is familiar from English: the EXT cannot be a pronoun coindexed with an R-expression in the INT.

(36) *Condition C: Non-absolute EXT > Non-absolute INT*

Na-kiring-ang a' [EXT *pro*] [INT buku-na iAli].
 3ERG-send-APPL 1ABS he book-3GEN NAME
 'He_{*i,j} sent me Ali_i's books.'

In the clauses where the INT is the absolutive DP, however, this pattern is reversed: the EXT *can* be a pronoun coindexed with an R-expression in the INT, like a possessor (37).

(37) *Condition C: Absolutive INT > Non-absolute EXT*

Na-waca i [EXT *pro*] [INT buku-na iAli] dionging.
 3ERG-send-APPL 3ABS he book-3GEN NAME yesterday
 'He_{i,j} read Ali_i's book yesterday.'

In the same context, moreover, the force of Condition C begins to cut the other way: the EXT loses the ability to contain R-expressions coindexed with a pronominal INT.

(38) *Condition C: Absolutive INT > Non-absolute EXT*

Na-peroa i [EXT sola-na iAli na iMina] [INT *pro*] dionging.
 3ERG-invite 3ABS friend-3GEN NAME and NAME him yesterday
 'Ali_i and Mina's friends invited him_{*i,j} yesterday.'

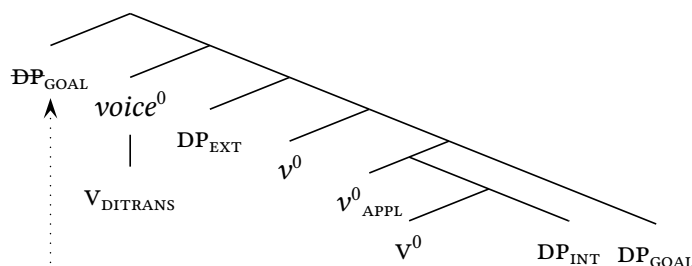
The force of these observations is to suggest, independent of the facts of linear order, that in the transitive voice the INT must raise to a position that c-commands the EXT. And this step of movement, in turn, is mirrored by a parallel process that targets the ditransitive GOAL. In the ditransitive voice, we have seen that EXT c-commands the non-absolute INT (36). But a different relationship holds between the EXT and the absolutive GOAL: the EXT can be a pronoun coindexed with an R-expression in the GOAL (39a) and it cannot contain an R-expression coindexed with a pronominal GOAL (39b).

(39) *Condition C: Ditransitive GOAL > EXT*

- a. Na-kiring-ang i [EXT *pro*] [INT buku] [GOAL sola-na iAli].
 3ERG-send-APPL 3ABS he book friend-3GEN NAME
 'He_{i,j} sent Ali_i's friends the book.'
- b. Na-kiring-ang i [EXT sola-na iAli] [INT buku] [GOAL *pro*].
 3ERG-send-APPL 3ABS friend-3GEN NAME book him
 'Ali_i's friends sent him_{*i,j} the book.'

It would thus seem that both the transitive INT and the ditransitive GOAL must raise, albeit usually covertly, to a position that C-commands the EXT. It thus seems eminently plausible to posit the rough A-syntax in (40): across all clauses of the language, the absolutive DP raises above the base position of the EXT and lands in SPEC, *voiceP*.

(40) *The Ditransitive GOAL: Raises above the EXT*



Working from a wider network of similar patterns, [Brodkin 2022b](#) proposes an analysis of Mandarin syntax with the same essential shape: the absolutive DP undergoes covert movement to the highest A-position in the clause. In broad perspective, this analysis is satisfying for the fact that it redeems a classical intuition that has underlied much work on Western Austronesian languages: the intuition that the absolutive DP is, in some sense, the structural subject. This position, made explicit in the earliest structuralist work on Tagalog ([Bloomfield, 1917](#)), has since been picked up by a major strand of generative work on Western Austronesian languages since [Guilfoyle et al. 1992](#) (Tagalog ([Guilfoyle et al., 1992](#); [Aldridge, 2004](#)), Malagasy ([Guilfoyle et al., 1992](#); [Ting, 2022, 2023](#)), Indonesian ([Aldridge, 2008](#)) and Malay ([Nomoto, 2013, 2015](#)), Acehnese ([Legate, 2014](#)), and Toba Batak ([Erlewine, 2018](#))). In wider perspective, it also establishes a productive parallel between these languages and the wider class of ergative languages that show High Absolutive syntax: systems where, in the tradition of [Bittner & Hale 1996a,b](#), the absolutive DP receives the highest clause-internal source of structural Case. In the context of our present investigation, this conclusion opens up a natural path to understand the syntax that lies beneath Absolutive Preposing.

3.2 The Position of the Absolutive DP

Building from the understanding that the absolutive DP always controls agreement on τ^0 (a position defended at length in that paper on both morphological and syntactic grounds), [Brodkin 2022b](#) proposes an account of Mandarin syntax on which the absolutive DP raises directly from a low position in the *vp* to SPEC,TP. But entirely within the syntax, there is

good reason to believe that the absolutive DP must transit through SPEC,*voiceP*—and the case for that view emerges from the syntax of one type of non-finite clause. Mandar has a preposition *mau* “though” that can select complement clauses of several different sizes, and one of these option is presented in (41): a non-finite clause that contains all the usual auxiliaries but not the usual pattern of absolutive agreement.

(41) *Non-Finite Complements of Though*

- a. **Mau** [_{POLP} ndang ____ [_{ASPP} rua [_{voiceP} na-ita dio di perpustakaan]]],
 though not ever 3ERG-see there in library
 ‘Though never seen in the library,’
- b. ... na-jamangang i de’ iAli skripsi-na.
 3ERG-work on 3ABS allegedly NAME thesis-3GEN
 ‘...Ali is allegedly working on his thesis.’

I propose that the absence of absolutive agreement reflects the following fact: *mau* selects a complement clause that is smaller than a TP. This fact is important for the following reason: the reduced complements of *mau* are able to host overt absolutive DPs, as in (42), and these elements, on this analysis, cannot possibly sit in SPEC,TP.

(42) *Non-Finite Complements of ‘Though’: ✓ Absolutive Arguments*

- Mau [_{voiceP} na-tulisi introduksi-na]],
 though 3ERG-write introduction-3GEN
 ‘Though he wrote [–FIN] the introduction,’

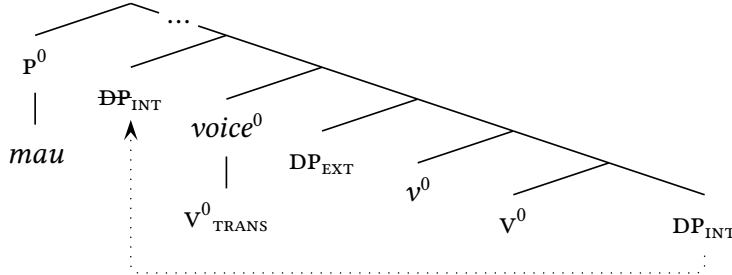
Despite this fact, the absolutive DPs that appear within these reduced clauses show the same privileges of binding as their counterparts in finite TPs. When the verb is transitive in a reduced clause introduced by *mau*, for instance, the facts of Condition C continue to suggest that the absolutive INT moves above the EXT. As a result, the INT can contain an R-expression that is coindexed with a pronominal EXT in this context (43a)—and it cannot be a pronoun that is coindexed with an R-expression in the EXT (43b).

(43) *Non-Finite Complements of ‘Though’: Absolutive Arguments Raise*

- a. Mau [_{voiceP} na-olo’i [_{EXT} *pro*] [_{INT} teori-na iAli]],
 though 3ERG-like he theory-3GEN NAME
 ‘Though he_i likes Ali’s_i theory,’
- b. Mau [_{voiceP} na-pakananyang [_{EXT} dospeng-u yau na iAli] [_{INT} *pro*]],
 though 3ERG-encourage advisor-1GEN me and NAME him
 ‘Though Ali’s_i and my advisor encouraged him_{*i},’

These facts suggest that the absolutive DP must raise to a high A-position in a domain smaller than the TP, targeting a position low enough to surface in the reduced clauses above. I propose that this low position is SPEC,voiceP. On this view, the clause in (42) shows the syntax in (44): when the verb is transitive, *voice*⁰ attracts the INT to SPEC,voiceP.

(44) *Absolutive DPs* → SPEC,voiceP



At a theory-internal level, the result is that we arrive at an analysis of Mandar that converges with a wide literature that takes the High Absolutive configuration to emerge from a step of A-movement to the edge of the clause-internal phase—both in Austronesian (Rackowski, 2002; Aldridge, 2004) and beyond (Coon *et al.*, 2014, 2021; Royer, 2022, 2023). More specifically, moreover, we arrive at an account that converges with the strand of the literature that takes this step of movement to target the specifier of *voice*⁰—here understood as the head that defines the clause-internal phase—and not the specifier of *v*⁰, the head that introduces the EXT (Nomoto, 2013, 2015; Erlewine & Sommerlot, To Appear).

At an empirical level, in turn, we arrive at syntactic confirmation for our initial theory of what Absolutive Preposing is: when the absolutive DP surfaces above the verb—as in the *mau*-clause in (45)—we can imagine that it undergoes its usual step of syntactic movement to SPEC,voiceP and is then forcibly spelled out in that position for reasons of the phonology.

(45) *Non-Finite Complements of ‘Though’: Absolutive Preposing*

Mau [POLP ndappa [ASP rua [voiceP penutupang na-pikkiri iAli ____]],
 though not.yet ever conclusion 3ERG-think of NAME
 ‘Though Ali hasn’t yet thought about [–FIN] the conclusion,’

Much work remains to be done to show that we can productively understand this system in this way. But we can nevertheless observe at the outset that the proposal in (??) fits naturally with the core architectural commitments of the minimalist program—surrounding the nature of movement and the nature of the relationship between phonology and syntax. To the extent that we can show that Absolutive Preposing must be understood along these lines, then, we can leverage this pattern to mount a case that this vision must be correct.

4 Prosodic Background

We thus embark on the second arc of this paper with the following analytical goal in hand: to weigh the narrow phonological factors that seem to trigger Absolute Preposing against the phonology of the language in search for a motive for this pattern of spell-out. Looking beyond this empirical horizon, moreover, we can also stake out the theoretical goals that we must meet to offer a satisfactory and meaningful analysis of this system.

(46) *A Reasonable Analysis of Absolute Preposing Must:*

- a. Explain why the phonology should force spell-out of the absolute DP in SPEC, *voiceP* when (i) that DP carries one instance of word-level stress, (ii) the remainder of the *voiceP* hosts two, and (iii) at least one auxiliary appears above the *voiceP*,
- b. Explain why the absolute DP is spelled out in its base position otherwise, and
- c. Situate its results in a reasonable theory of the Syntax-Phonology Interface.

To this end, we can begin by identifying the type of phonological pressure that could plausibly drive our effect: one that prevented the absolute DP from being spelled out in its base position in the exact phonological contexts where absolute preposing is forced. As a first pass, we can begin with the descriptive constraint in (47).

(47) *The Driving Constraint: First Pass*

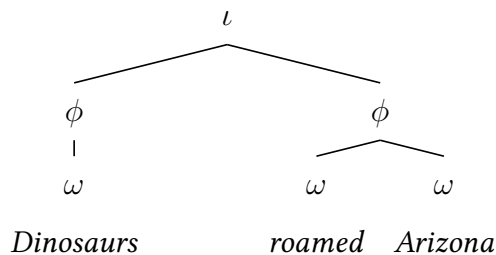
When the absolute DP carries one stress, the remainder of the *voiceP* hosts two, and at least one auxiliary appears above the *voiceP*, some phonological pressure must prevent the absolute DP from being spelled out beneath SPEC, *voiceP*.

This initial constraint cannot be the end of our analysis, because it refers to various types of syntactic information—like the category of auxiliary and the label of the position SPEC, *voiceP*—in a way that is standardly held to be impossible for phonological constraints (the principle of SYNTAX-FREE PHONOLOGY: Nespor & Vogel 1986). But it is useful for the way that it directs our attention to the types of genuine phonological pressures that might drive this type of effect: those that govern the output shapes of constituents like the *voiceP*. These are the constraints that guide the construction of phrase-level phonological domains in the output system of prosodic organization—and in particular, those that enforce requirements of phonological balance within those domains (Prieto, 2005; Elordieta, 2006; Itô & Mester, 2007). The first step of our next arc, then, will be to establish the scaffolding to understand how these pressures might operate in the phrasal phonology of Mandarin.

4.1 Detecting Prosodic Constituents

The theory of the Prosodic Hierarchy holds that phonological strings are also organized into hierarchical constituent structures above the metrical level of syllables and feet (Selkirk, 1978, 1984, 1986; Nespor & Vogel, 1986). In the minimal and universal framework of Itô & Mester 2007, this suprametrical constituent structure is built from exactly three kinds of abstract phonological constituents: the prosodic word (ω), the phonological phrase (ϕ), and the intonational phrase (ι). Example (48) shows the way that sentences are parsed into prosodic structures built from these constituents: in the sentence *Dinosaurs roamed Arizona*, the subject, verb, and object each form ω s, the subject forms a ϕ and so too does the sequence of verb and object, and both of those ϕ s are then parsed into a single ι .

(48) *Phonological Organization above the Word*



To see how this type of structure guides the system of Absolute Preposing, it will be useful to review the Mandarin-internal diagnostics that have steadily amassed to detect the edges of specific set of prosodic domains that underlie the core effect in Mandarin. Our study begins with the ω and ϕ , which can be detected from the distribution of stresses and tones (Brodkin, 2025). In Mandarin, the ω hosts a single instance of word-level stress, which falls on the penult and correlates with a low tone (Pelenkahu *et al.*, 1983).² The ϕ , in turn, carries a single high tone (H) at its right edge. The following example shows the parse that emerges from these diagnostics in the DP *itim buku cennirrara kaiang* “that big book of love spells”: the facts of stress shows that *buku* ‘book’, *cennirrara* ‘love spell’, and *kaiang* ‘big’ each form ω s, and the H-tone shows that the full DP forms a ϕ .

²Word-level stress falls on every N⁰, V⁰, ADJ⁰, and ADV⁰ in Mandarin, including many elements that surface in second-position. It is absent from the functional x⁰s P⁰, C⁰, POL⁰, and ASP⁰ (which hosts auxiliaries): a natural distribution if only the lexical categories of Truckenbrodt 1999 are mapped to ω s (McCarthy & Prince, 1993; Selkirk, 1995, 2009). Brodkin 2025 argues that the distribution of word-level stress arises from the right-alignment of a single trochee in the ω , as proposed in other languages of the subfamily (Mithun & Basri, 1986; Friberg & Friberg, 1991; Broselow, 1999; Basri *et al.*, 1999, 2012; Jukes, 2006)

(49) *Diagnostics for the Word and Phrase*

wátʃa ^H	i	itim búku tʃennirrara káiaŋ ^H
u-baca	i	itiŋ buku tʃennirrara kaiaŋ ^H
1ERG-read 3ABS		that book love spell big

‘I read that big book of love spells.’

To build toward an understanding of Absolutive Preposing, we will also have to understand the ways in which the smallest ϕ s in the language are organized into successively larger prosodic constituents. To this end, we will leverage a set of phonotactic restrictions in Mandar, documented in (Brodkin, To Appear), that converge to reveal the ways in which minimal ϕ s are parsed together into larger ϕ s beneath the level of the intonational phrase.³ The restrictions that will guide us in this endeavor are the following four.

FIRST: Mandar has a process of Coalescence that reduces the sequences /ai ae/ to [e] and /au ao/ to [o]. The following example shows how this process operates in three-word vsx clauses. When the verb ends in a vowel sequence in this context, it must show coalescence (/naitai/ “seek” → [néte]). But when the s ends in a final vowel sequence in this clause, it cannot undergo the same type of coalescence (/balao/ “mouse” → [wálo]). The x, too, is not able to show coalescence in this clause (/di bangae/ → [di wáŋge]).

(50) *Coalescence: The First Asymmetry*

néte ^H	i	waláo ^H	di wangáe ^H
na-itai	i	balao	di bangae
3ERG-look for	3ABS	mouse	in PLACE

‘The mouse looked for it in Banggae.’

³These restrictions are easy to hear in the spoken language, and they are typically reflected in the popular orthography, but they are often enforced imperfectly in regular speech—as both patterns of phonological phrasing and processes in the phrasal phonology can be derailed by the phonology of focus and by a wide range of non-phonological factors, including speech rate and planning constraints. As a result, our discussion will focus on the ways in which speakers judge these restrictions to emerge in well-planned clauses that are produced at a regular speech rate under broad focus. The judgments reported below have been gathered over three months of targeted work with Jupri Talib, with whom I have been working for six years overall (and for four years on prosodic phonology). To collect each judgment, Jupri was given a pair of written sentences that differed in the application of a phonological rule. Once the sentences were established as pre-phonologically acceptable, he was then walked through a context in which the sentences could be uttered under broad focus and asked to produce each sentence several times. Once the sentences had become familiar, Jupri was then asked whether one version could be identified as acceptable and comparatively “more normal” (Indonesian: *lebih biasa*) in contexts that of broad focus (*konteks biasa*) and then asked to rate the second example as “also possible with regular prosody” (*bisa juga dengan irama biasa*), “only possible with narrow focus” (*hanya dengan penekanan khusus*), or “incorrect” (*salah*).

SECOND: Mandar has a process of Gliding that reduces prevocalic /i u/ to [j w]. In three-word vsx clauses, it applies at the right edge of the v but not at the right edges of the s or x. Thus in example (51), the v /napapia/ “make” reduces to [napápja], but the s /imaria/ “NAME” cannot reduce to [imárja] and the x /lamasarian/ cannot reduce to [lamasárja].

(51) *Gliding: A Second Asymmetry*

napápja ^H	i	imaría ^H	di lamasaríanj ^H
na-papia	i	imaria	di lamasarianj
3ERG-make	3ABS	NAME	in PLACE

‘Maria made it in Lamasariang.’

THIRD: Mandar has a process that deletes intervocalic /ʔ/ (which is always parsed as a coda) and intervocalic coda /ŋ/ (which is parsed as a coda word-finally). I will refer to this process as Glottal Coda Deletion, building from the understanding (well-motivated in Mandar, I believe), that /ŋ/ is phonologically a glottal segment (De Lacy, 2006). This process has the same distribution as Coalescence and Gliding: in a three-word vsx clause, it applies in the v to reduce /nakaraʔus/, “scratch” to [nakáros] (with Glottal Coda Deletion feeding Coalescence). But it is blocked at the right edge of the s, /iripaʔi/ (a NAME), which cannot reduce to [irípe], and the x /rebataʔa/, which cannot reduce to [rewáta].

(52) *Glottal Coda Deletion: A Third Asymmetry*

nakáros ^H	i	iripáʔi ^H	di rewatáʔa ^H
na-karaʔus	i	iripaʔi	di rebataʔa
3ERG-scratch	3ABS	NAME	in PLACE

‘Ripa’i scratched it in Rebata’a.’

FOURTH: Mandar has a process of Voiced Obstruent Lenition that lenites intervocalic /b d d̥ ʒ g/ to [w ɹ j ɣ]. At the junctures that require Coalescence, Gliding, and Glottal Coda Deletion at the right edge of one ϕ, this process applies at the left edge of the next. In a three-word vsx clause, then, Voiced Obstruent Lenition must apply at left edge of the s when the preceding segment is a vowel: thus the s /do guru/ “that teacher” is reduced to [ɹo yúru] (53). Voiced Obstruent Lenition is then blocked at the left edge of the x, such that the x /di genaʔ/ “earlier” cannot be realized as [ɹi yénaʔ].

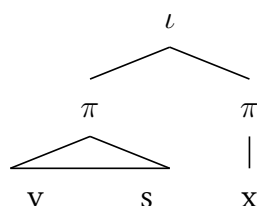
(53) *Voiced Obstruent Lenition: A Fourth Asymmetry*

nawátʃa ^H	i	ɹo yúru ^H	di yénaʔ ^H	ó ^H
na-batʃa	i	do guru	di genaʔ	o
3ERG-read	3ABS	that teacher	in earlier	there

‘That teacher there read it earlier.’

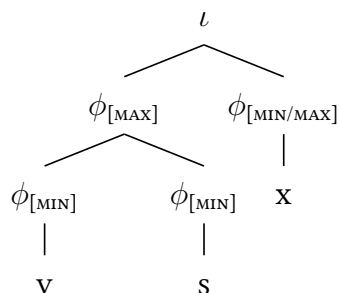
Brodkin To Appear argues that these restrictions reveal the prosodic constituency in diagram (54): in three-word vsx clauses, the vs string forms a prosodic constituent π that allows Coalescence, Gliding, Glottal Coda Deletion, and Voiced Obstruent Lenition internally. The right edge of this domain blocks the first three processes, banning final Coalescence, Gliding, and Glottal Coda Deletion in the s and x. The left edge of this domain then blocks the last process, banning initial Voiced Obstruent Lenition in the x.

(54) *The VSX String: Constituency*



There is a natural candidate for a constituent in this position, and it is the maximal ϕ : namely, the type of ϕ that is directly dominated by the ι (Itô & Mester, 2007, 2012, 2013). The following diagram shows the shape of this revised analysis, which shows a pattern of recursive prosodic phrasing: the v, s, and x each form a minimal ϕ (seen from the distribution of H-tones), and the minimal ϕ s that contain the v and the s are then parsed into another higher ϕ . This higher ϕ is directly dominated by the ι , and as a result, it acquires the relational property of maximality, marked with the subscripted [MAX]. The minimal ϕ that contains the x is also rendered maximal for this reason. It is the edges of the ϕ s that are maximal, on this analysis, that block the phonological processes above.

(55) **Brodkin To Appear:** *The Maximal ϕ*



4.2 Prosodic Problems in the VoiceP

Armed with these diagnostics for prosodic constituency, we can now see that there is, in fact, a phonological problem that emerges in the contexts that force Absolutive Preposing.

To understand it, we can begin with the baseline prosody of the Mandarin *voiceP*. In the verb-initial clauses of the language, **Brodkin To Appear** shows that the *voiceP* is generally parsed into a single maximal ϕ . The string vso, for instance, is always parsed into a single ϕ_{MAX} when the v, s, and o each form a single ω (=when each carries one stress). We can see this fact in clauses like (56), where a three-word vso string is followed by a *voiceP*-final adjunct like *digena* ‘earlier.’ In that context, all of our processes are forced to apply between the v, s, and o but cannot apply between the o and the following x. At the left edge of the x, then, Voiced Obstruent Lenition, then, is blocked (*/digena?/* $\nrightarrow$ [*ɿyena?*]).

(56) *Phonological Diagnostics: Reveal that the voiceP* $\rightarrow \phi_{\text{MAX}}$

{ ϕ_{MAX} }				{ ϕ_{MAX} }	{ ϕ_{MAX} }	
nakiríŋa ^H	õ	ɿo	rípe ^H	wúku ^H	ɗiyéna? ^H	ó ^H
na-kiriŋ-aŋ	o	do	ripaʔi	buku	digenaʔ	o
3ERG-send-APPL	2ABS	that	NAME	book	earlier	there

‘That Ripa’i there sent you a book earlier.’

The parse is the same whenever the verb and the arguments in the *voiceP* collectively amount to three or fewer prosodic ω s. For instance, the string vso receives the same parse when the verb is antipassive and its prefix takes the shape *m*- (57a). The same parse also emerges in three- ω strings of similar orders like vsd and vod with ditransitive verbs (57b). In all cases, moreover, it is unaffected by the introduction of final adjuncts to the *voiceP* (following known generalizations on the prosody of adjunction: **Brodkin To Appear**).

(57) *Every Three-Word VoicePs of the order VSOD* $\rightarrow \phi_{\text{MAX}}$

- a. { ϕ_{MAX} } { ϕ_{MAX} }
- | | | | | |
|--------------------|------|-------------------|--------------------|------------------|
| mánde ^H | i | jóna ^H | jólen ^H | dío ^H |
| maŋ-ande | i | ɗʒoŋa | ɗʒolen | dio |
| ANTIP-eat | 3ABS | deer | guava | there |
- ‘The deer are eating guavas there.’
- b. { ϕ_{MAX} } { ϕ_{MAX} }
- | | | | | |
|----------------------|------|--------------------|-------------------|---------------------|
| najolóa ^H | ĩ | irípe ^H | yúru ^H | ɗjóniŋ ^H |
| na-ɗʒolo-aŋ | i | iripaʔi | guru | dioniŋ |
| 3ERG-point-APPL | 3ABS | NAME | teacher | yesterday |
- ‘Ripa’i showed it to the teacher yesterday.’

The parse of *voiceP*-final adjuncts, in turn, is stable as well: these elements invariably fall outside the ϕ_{MAX} that corresponds to the *voiceP*, and each adjunct forms a ϕ_{MAX} of its own. This parse holds even when the verb and its arguments amount to less than three

prosodic ω s: for instance, when a vs string is followed by the adjuncts *dio* “there,” *dionging* “yesterday,” and *o* “there,” the parse must be $\{\phi_{[MAX]} \text{ vs} \} \{\phi_{[MAX]} \text{ x} \} \{\phi_{[MAX]} \text{ x} \} \{\phi_{[MAX]} \text{ x} \}$.

(58) *VSX Clauses: Each Adjunct = ϕ_{MAX}*

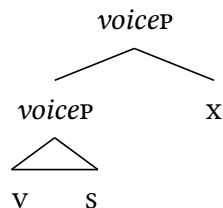
$\{\phi_{[MAX]}$				$\}$	$\{\phi_{[MAX]}$	$\}$	$\{\phi_{[MAX]}$	$\}$	$\{\phi_{[MAX]}$	$\}$
nasáka ^H	i	ɔo	waláo ^H		dío ^H		djónɿŋ ^H		ó ^H	
na-saka	i	do	balao		dio		dionɿŋ		o	
3ERG-catch	3ABS	that	mouse		there		yesterday		there	

‘That mouse caught it there yesterday.’

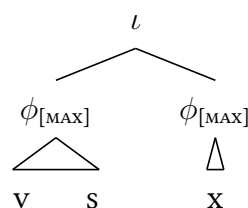
We thus arrive at two generalizations of a familiar shape. First, in the default case, the prosody mirrors the syntax beneath it, and thus xps like the *voiceP* are typically parsed into ϕ s. Second, adjunction structures receive a consistent and special prosodic parse: adjuncts are prosodically set off from their hosts (Truckenbrodt 1999; also Chen 1987; Hale & Selkirk 1987) and required, in Mandar, to form independent maximal ϕ s.

(59) *The Basics of Prosodic Phrasing*

a. Syntax



b. Prosody



With this much in place, we turn to the matter of weight. Problems emerge in Mandar when the verb and the arguments in the *voiceP* collectively amount to more than three prosodic ω s. When the verb in example (56) is reduplicated, for instance, the distribution of our phonological processes is forced to change: in vso strings of this type, Glottal Coda Deletion and Voiced Obstruent Lenition cannot apply between the v and the s.

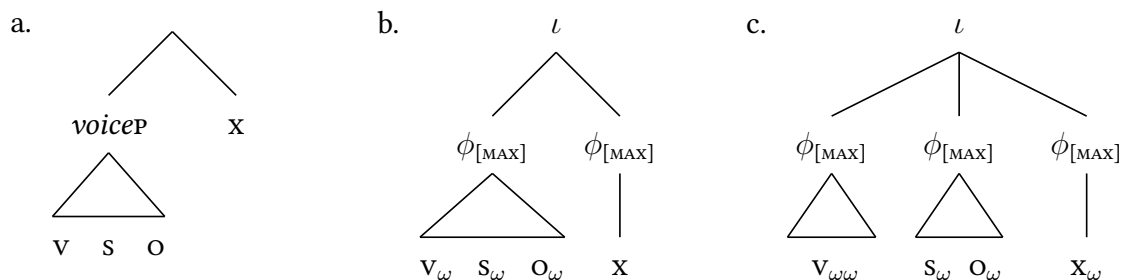
(60) *Phonological Diagnostics: Reveal that the voiceP $\rightarrow \phi_{MAX}$*

$\{\phi_{[MAX]}$				$\}$	$\{\phi_{[MAX]}$			$\}$	$\{\phi_{[MAX]}$			$\}$
náki-nakirɿŋaŋ ^H	o	do	rípe ^H	wúku ^H	diyénaʔ ^H		ó ^H					
óσ-na-kirɿŋ-aŋ	o	do	ripaʔi	buku	digenaʔ		o					
RED-3ERG-send-APPL	2ABS	that	NAME	book	earlier		there					

‘That Ripa’i there sent you a few books earlier.’

These phonological changes suggest that the introduction of an additional prosodic word into the *voiceP* triggers the following shift: the *voiceP* in (61a) is parsed into one ϕ_{MAX} when it consists of three words (61b), but it is split into two when it consists of four (61c).

(61) *Weight Constraints in the voiceP*



We thus arrive at another pattern familiar from the literature on prosodic organization: the usual symmetry between the prosody and the syntax begins to break down when more and more ω s are added into xps. In the terminology of Optimality Theory (Prince & Smolensky, 1993/2004), we can understand this pattern to emerge from the ranking of two constraints: a violable interface pressure that demands xps to form ϕ s (62a) and a higher-ranking phonological requirement that prohibits ϕ s of more than three ω s (62b).

(62) *Parsing the Mandarin voiceP: Starting Principles*

a. *Syntax-Prosody Isomorphism*

Every xp in the input syntax must correspond to a ϕ in the output prosody.

b. *Phonology-Internal Balance*

Output ϕ s in Mandarin cannot contain more than three ω s.

In Mandarin, Brodtkin To Appear shows that this second requirement can be understood to emerge from the interaction of two familiar types of output-oriented constraints. These are the constraints that demand strict binary branching in prosodic structure of the prosody (Prince 1980; Itô & Mester 1992, and especially Kubozono 1989; Selkirk 2000), and the relevant subcases of these constraints are presented below. The first of the pair, MINIMAL BINARITY, demands that every ϕ_{MAX} contain two daughters and no fewer; the second, MAXIMAL BINARITY, demands that every ϕ_{MAX} contain two daughters and no more.

(63) *Parsing the Mandarin voiceP: The Constraints on Weight*

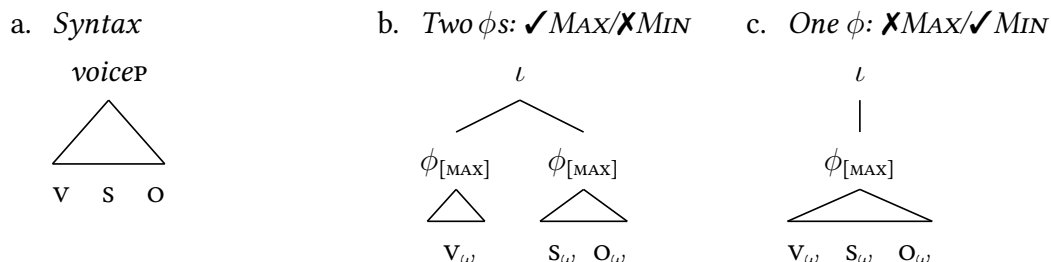
a. *Minimal Binarity*: every ϕ_{MAX} must contain at least two daughters.

b. *Maximal Binarity*: every ϕ_{MAX} must contain at most two daughters.

With the right ranking, these constraints will allow us to derive the restriction in (62b): the language must allow the ϕ_{MAX} to contain three ω s but not four. This restriction emerges naturally from the ranking of MINIMAL BINARITY over MAXIMAL BINARITY, which has

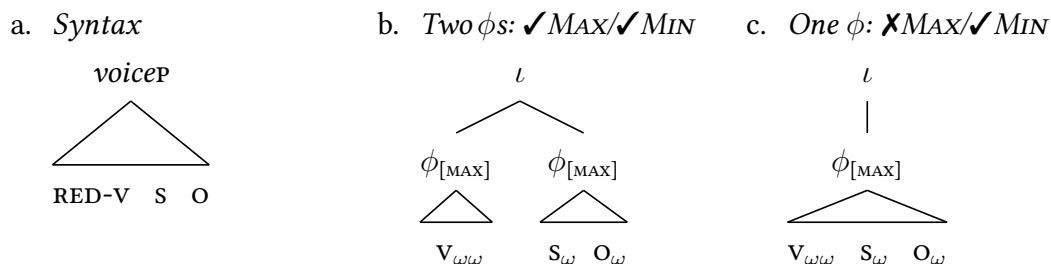
the following effect. In a derivation where the *voiceP* contains exactly three words, the constraint MAXIMAL BINARITY will favor the parse in (64b), where the *voiceP* is split up into two maximal ϕ s that each contain no more than two ω s. But this type of parse will always violate MINIMAL BINARITY, which prohibits maximal ϕ s that contain only one ω . The ranking of MINIMAL BINARITY over MAXIMAL BINARITY will thus force the attested parse in (64c)—where the *voiceP* is parsed into a single ϕ_{MAX} that violates MAXIMAL BINARITY.

(64) *The Three-Word Condition: Minimal Binarity > Maximal Binarity*



In the contexts where the *voiceP* contains four ω s, this same system will allow us to derive the attested parse if we posit—informally for now—that MAXIMAL BINARITY outranks SYNTAX-PROSODY ISOMORPHISM. In this context, MAXIMAL BINARITY will make its usual demand: the *voiceP* must be parsed into two maximal ϕ s. But here the ensuing parse—in (65b)—will not violate MINIMAL BINARITY, as each maximal ϕ s contains two ω s. MAXIMAL BINARITY alone can thus force this parse over the integrated alternative in (65c).

(65) *The Four-Word Condition: Maximal Binarity > Syntax-Prosody Isomorphism*



4.3 Minimal Binarity and Absolutive Preposing

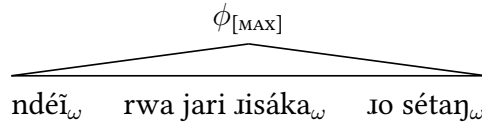
We thus return to Absolutive Preposing with two new facts in hand: (i) the verb and the arguments in the Mandar *voiceP* are typically parsed into a ϕ_{MAX} and (ii) MINIMAL BINARITY, which bans one- ω maximal ϕ s, is a high-ranked and active pressure in the language.

It is against this backdrop that we turn to the prosody of clauses that host auxiliaries. In Mandar, the preverbal auxiliaries do not form independent prosodic words (on a par with most functional heads in Mandar (Brodkin, 2025) and elsewhere (Selkirk, 1995; Itô & Mester, 2009)) when they are not followed by absolutive agreement (for instance, in fragments and non-finite clauses). But in the finite clauses of the language, the auxiliary that hosts absolutive agreement—in example (66), the negator *ndang*—is forced to form its own phonological phrase. This shift is forced by the particular phonology of the absolutive agreement enclitics, and its local effect is to subject the auxiliary to the phonology that emerges at the right edge of the minimal ϕ (Brodkin 2025)—a topic sidestepped here.

- (66) *The Auxiliary that hosts Absolutive Agreement* $\rightarrow \phi$
- | | | | |
|-------------------------|------------------------------|-------------------------|----------------|
| $\{\phi_{[\text{MAX}]}$ | } | $\{\phi_{[\text{MAX}]}$ | } |
| ndéĩ ^H | rwa jari .isáka ^H | .ɔ sɛtaŋ ^H | ó ^H |
| ndaŋ i | rua jari di-saka | do setaŋ | o |
| not | 3ABS ever DID PASS-catch | that demon | there |
- ‘That demon there never WAS caught.’

Within the present investigation, this fact is important for the following reason: when the highest auxiliary is forced to form a ϕ , it is also forced to form its own prosodic word. This shift is driven by the requirement for HEADEDNESS (Nespor & Vogel, 1986), which demands that each prosodic constituent dominate a constituent of equal or immediately smaller size on the prosodic hierarchy (such that every ϕ must contain a ω or a smaller ϕ). Within the target auxiliary, this type of promotion is visible in the distribution of word-level stress, which also falls on the auxiliary that hosts absolutive agreement (66). But it also has an effect on the higher-level prosody of the clause: in a finite AVS string like (66), the string AUX-ABS.CLITIC-OTHER.AUXES-V-S is parsed into a ϕ_{MAX} that contains three ω s.

- (67) *Aux-Verb-Subject Clauses: Three-word ϕ_{MAX}*



With this much in place, we turn to the parse that emerges in the contexts that force Absolutive Preposing: descriptively speaking, those clauses that contain an auxiliary, a following absolutive enclitic, and a *voiceP* that contains three ω s. It is natural to begin in this connection with a clause where Absolutive Preposing cannot apply, so that we can

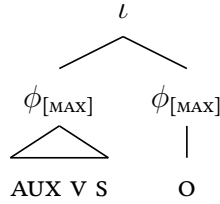
see the type of parse that is avoided in the cases where it occurs. This parse is shown in the example in (68), which presents the prosody of a ditransitive clause of the order AVSO. There, the verb is followed by two non-absolutive arguments that each form a single ω .

(68) *The Parse of Aux-Verb-DP-DP*

$\{\phi_{[MAX]}$				$\}$	$\{\phi_{[MAX]}$	$\}$
rwéi ^H	nállja ^H	írípáʔi ^H	búnja ^H			
rua i	na-alli-an	iripaʔi	buŋa			
once 3ABS	3ERG-buy-APPL	NAME	flower			
'Ripa'i once bought her flowers.'						

The parse that emerges in this context is presented in tree (69). There are a number of ways in which this parse is surprising, but what matters at present is only this: when an auxiliary and verb are followed by two ω -sized DPs, the second DP forms a one- ω ϕ_{MAX} .

(69) *The Parse of Aux-Verb-DP-DP*



We are now ready to see the parse that emerges when we replace one of the DPs in (69) with a ω -sized absolutive DP. This is shown in (70), which is a ditransitive clause where the GOAL is the name *Ripa'i*. In this context, the GOAL must undergo Absolutive Preposing, and the prosodic effect of that operation is clear: it eliminates the configuration in (69) and yields a parse without single- ω maximal ϕ s: $\{\phi_{[MAX]} \text{ AUX}_\omega \text{ DP}_\omega\} \{\phi_{[MAX]} \text{ V}_\omega \text{ DP}_\omega\}$.

(70) *The Parse of Aux-DP-Verb-DP*

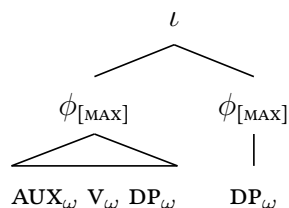
$\{\phi_{[MAX]}$		$\}$	$\{\phi_{[MAX]}$	$\}$
rwéi ^H	írípáʔi ^H	wállja ^H	ánde ^H	—
rua i	iripaʔi	u-alli-an	ande	
once 3ABS	NAME	1ERG-buy-APPL	food	
'I once bought Ripa'i food.'				

In this context, then, Absolutive Preposing has a clear effect on the surface prosody. The systems that guide the syntax-prosody mapping canonically parse strings of the order AUX-V-DP-DP into prosodic structures of the shape $\{\phi_{[MAX]} \text{ AUX V DP}\} \{\phi_{[MAX]} \text{ DP}\}$, violating

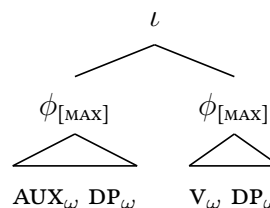
Minimal Binarity (71a). But the same systems will always parse strings of the shape AUX-DP-V-DP as $\{\phi_{[MAX]} \text{ AUX DP} \} \{ \phi_{[MAX]} \text{ V DP} \}$ —yielding no violations of that constraint (71b).

(71) *The Surface Effect of Absolutive Preposing*

a. *Base Order: ✗ Minimal Binarity*



b. *Preposing: ✓ Minimal Binarity*



This advance lets us restate the descriptive effect of Absolutive Preposing as follows.

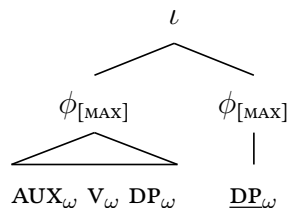
(72) *Absolutive Preposing: The Final Description*

In clauses that host auxiliaries, the absolutive DP is spelled out in SPEC, *voiceP* when that linearization would avoid a violation of Minimal Binarity that would emerge if the absolutive DP were spelled out in its base position.

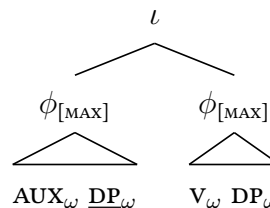
This analysis captures the shape of Absolutive Preposing in an elegant and integrated way, and in wider perspective, it is important for the following fact: it commits us to an architectural vision on which the linearization of movement dependencies must be able to be influenced by output phonological constraints. The system that linearizes movement dependencies, more specifically, must be able to see the clause-sized prosodic parses that would be built around different patterns of linearization and must weigh them against output constraints to determine the exact positions in which specific DPs will be spelled out. The result is a system that must compare global phonological structures like (71a)-(71b) to determine whether the absolutive DP will be spelled out in SPEC, *voiceP*.

(73) *The Position of Spell-Out: Determined with respect to Output Prosody*

a. *Low Spell-Out: ✗ Minimal Binarity*



b. *High Spell-Out: ✓ Minimal Binarity*



This conclusion establishes a central line of convergence between the present study of Mandarin and the many strands of research on phonologically-triggered spell-out of moved elements in unexpected positions along their paths. In Romanian, for instance, Bošković 2002 observes that all WH-phrases typically front to the left periphery (74a) but that non-subject WH-phrases are spelled out low if homophonous with moved WH-subjects (74b).

(74) *Phonological Constraints guide the Linearization of Movement Chains*

- a. Cine **ce** precede **ee**?
 who what precedes
 ‘Who precedes what?’
- b. Ce **ee** precede **ce**?
 What what precedes what
 ‘What precedes what?’

Bošković 2002 argues for an analysis that follows the logic above: the lower WH-phrase moves in the syntax, the phonology inspects the outputs created by all possible patterns of linearization, and it selects the output in (74b) as the phonologically optimal outcome.

The phonological factor that conditions this initial pattern of spell-out is relatively local, in that the choice of spell-out is determined by the shape of material adjacent to one position of the lower WH-phrase. But Franks 1998; Franks & King 2000 and Bošković 2001 argue that the system of linearization must also be able to reference the type of global prosodic structure that we have argued to condition our effect in Mandarin as well. Franks 1998; Franks & King 2000 and Bošković 2001 draw evidence for this type of interaction from Serbo-Croatian, where they argue that that it emerges in the second-position system. Franks & King 2000, for instance, propose that the string in (75a)—where an initial verb is followed by pronouns and auxiliaries—has the syntax in (75b): the pronouns and auxiliaries raise above the verb, but they are spelled out in their base positions in response to a constraint that bans them from appearing at the left edge of the intonational phrase

(75) *Global Phonological Constraints guide the Linearization of Movement Chains*

- a. Predstavio sam mu se
 introduced am to him REFLEXIVE
 ‘I introduced myself to him.’ Franks & King 2000 (reglossed)
- b. [_{FP} SAM MU SE [Predstavio sam mu se [PREDSTAVIO]]]
 am to him REFLEXIVE introduced
 ‘I introduced myself to him.’ Franks & King 2000 (reglossed)

The force of this connection is to deliver a valuable kind of scaffolding as we prepare to approach the final analytical desiderata that we staked out at the start of this section:

- (76) *Analytical Desideratum Three:* Repeated from (46c)
 [A reasonable analysis must] situate its results in a reasonable theory of the Syntax-Phonology Interface.

insofar as it establishes that the shape of our pattern in Mandar is not entirely unique: it turns on a type of interaction between clause-level prosodic phrasing and linearization that is relatively well-attested beyond the system under investigation above. The task that falls to us below, then, is to work out a theory of how exactly this interaction might occur.

5 Linearization and the Syntax-Prosody Interface

To understand how output constraints might guide these patterns of spell-out, it will be useful to begin with the structure of the phonological evaluation where prosodic structure is built. It has been widely recognized since the earliest work on lexical phonology (Kiparsky, 1982) that there is no evidence for cyclicity in phrasal phonology or in the construction of prosodic domains. This is the architectural position adopted by Match Theory (Selkirk, 2009, 2011), the leading theory of the interface between syntax and prosody, and it has been taken up by the vast majority of recent work at that interface (Itô & Mester 2012, 2013, 2019; Elfner 2012, 2015; Myrberg 2013; Kügler 2015; Elordieta 2015; Bennett *et al.* 2016, 2019; Hamlaoui & Szendrői 2017; Kalivoda 2018; Ishihara & Myrberg 2023; Brodtkin To Appear; for overviews, see Bennett & Elfner 2018, Ishihara & Kalivoda 2022).⁴

⁴There is a second tradition of work, beginning from the theory of phase-based spell-out proposed by Chomsky 2001, which attempts to derive surface patterns of phonological phrasing from phasal spell-out (Dobashi, 2003; Ishihara, 2004; Kahnemuyipour, 2005, 2009; Kratzer & Selkirk, 2007; Pak, 2008). The arguments put forward for this type of architecture have been almost fully theory-internal, and for two reasons its prospects are not good: (i) the empirical effects that have been argued to provide evidence for cyclicity can all be handled naturally in the fully parallel architecture of Match Theory (see Cheng & Downing 2016) and (ii) the shapes of phonological constituents are famously sensitive to their positions within higher layers of prosodic structure, such that the boundaries of constituents like the ω and ϕ often cannot be determined until their positions within the ι are known (Prieto, 2005; Elordieta, 2006; Itô & Mester, 2007). These interactions fall together naturally on the parallel theory of prosodic structure-building assumed in Match Theory, where all levels of prosodic constituency are built in a single round of phonological evaluation. On a cyclic model, however, they would require the edges of small prosodic domains to undergo continual types of derivational readjustment as higher levels of prosodic structure were introduced (a position put forward by Ackema & Neeleman 2003 and Bye & Svenonius 2012). If this view were correct, we might expect to see some kind of empirical evidence for this type of readjustment (e.g., opacity); to date, little has been found.

5.1 The Matter of Linearization

Following the standard theory of the syntax-prosody interface (Selkirk, 2009, 2011), then, I will assume that all prosodic structure is built within a single round of surface-oriented phonological evaluation that plays out over the entire output of the morphosyntactic derivation. On a strictly modular and feed-forward theory of the grammar, we can identify the input to this parallel and global phonological derivation as the output of the grammatical module that falls immediately before the phonology. Assuming an architecture on which the output of the syntax is passed through a distinct morphological module, as in Distributed Morphology (Halle & Marantz, 1993), we can thus identify the input to the phonology as the representation that is produced at the end of the morphological module.

This morphosyntactic representation is one that must have the following properties. Given that the mapping from syntax to prosody must access syntactic constituency and reflect syntactic relationships like selection and adjunction, this input representation must retain something like the hierarchical constituent structure of the syntax and some information about the syntactic relationships that hold between elements therein. But on the view that at least some type of linear order is established before the complete derivation is passed off to the phonology, this input must also contain a linear ordering of its terminal nodes: one that is plausibly established through the types of linearization algorithms developed by Embick & Noyer 2001, 2007; Embick 2007; Pak 2008, and Arregi & Nevins 2012 (though formulated in such a way as not to “flatten out” the constituency of the syntax).

For the purposes of this paper, it will suffice to assume that this process of linearization proceeds in the following way. Assuming the rough architecture of Arregi & Nevins 2012 and setting aside the matter of the cycle, there is a stage within the morphological module where linear relationships are imposed upon the terminal nodes of the syntax. These linear relationships are established by a system that (i) inspects the daughters of each node in the syntax, (ii) imposes an order between the two daughters (e.g., HEAD > COMPLEMENT or COMPLEMENT > HEAD) and then (iii) imposes an order between all the daughters one of daughter and all the daughters of the other (the Concatenation of Embick & Noyer 2001). The linearization that emerges from this system need not be final, as it now seems clear that the phonology must also be able to force movements of its own (Halpern, 1995; Chung, 2003; Bennett *et al.*, 2016; Aissen, 2017; Kusmer, 2020). But it must establish clear and unambiguous relative positions for each of the terminal nodes of the morphosyntax.

It is against this backdrop that we turn to the linearization of movement dependencies.

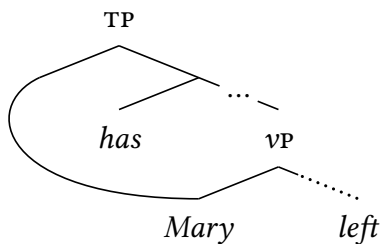
In the context of Minimalism, [Chomsky 1993](#) proposes that syntactic movements proceed in four steps: a moving element is first copied, the copy is then merged into a higher position, a syntactic object called a CHAIN is built around the two copies, and once the syntactic derivation is shipped off to PF, one copy is erased. The question that emerges on this analysis is thus the following: what mechanisms ensure that only one copy in each chain is pronounced at PF, favoring patterns of spell-out like (77a) over that in (77b)?

(77) *Chain Reduction: How and Why?*

- a. [_{TP} Mary has [_{vP} ~~Mary~~ left]].
- b. [_{TP} Mary has [_{vP} Mary left]].

The literature has put forward a range of solutions to this issue that preserve the basic assumption that chains and copies exist ([Bobaljik, 2002](#); [Fanselow, 2002](#); [Nunes, 2004](#); [Landau, 2006](#); [Corver & Nunes, 2007](#)). But there is a second strand of thought, laid out by [Gärtner 1999](#) and [Kracht 2001](#), that circumvents the issue by rejecting the postulation of chains and copies entirely—building in part from [Chomsky 1981](#). This is a view that takes movement dependencies to have the shape in (78): whenever an element seems to move from one position to another, it is simply merged directly into its derived position without being removed from the first. The result is that the moving element must be dominated by multiple nodes at once. As such, this theory is termed the theory of MULTIDOMINANCE.

(78) *Multidominance*



This reformulation provides a natural solution to the question of why moved elements are only realized in one place: there is only a single element to spell out. The result is a theory that yields a meaningful simplification to the structure of spell-out (along with additional gains: [Frampton 2004](#); [Johnson 2007](#); [Sauerland 2007](#)). Rather than postulate chains, copies, and chain reduction, this theory requires spell-out to execute exactly two tasks: (i) to ensure that individual elements are spelled out in exactly one position (yielding the type of one-to-one mapping that is typical of all input-output relationships in phonology), and (ii) to ensure that individual elements are realized in the highest positions they occupy (deriving the descriptive preference for what is generally termed “higher-copy spell-out”).

5.2 Chain Reduction at the Syntax-Prosody Interface

Within the architectural context laid out so far, we can understand this process to proceed as follows. At the end of the syntactic derivation that yields (77), the hierarchical structure in (78) is passed off to the morphological module. At the stage in that module where linearization applies, in turn, the result will be (79): a morphosyntactic representation that contains both hierarchical structure and linear order in which a single syntactic element—here the DP *Mary*—is simultaneously linearized in two distinct positions at once.

- (79) *The Output of the Morphosyntax*
 $[_{TP} \text{ Mary} > \text{ has} > [_{vP} \text{ Mary} > \text{ left}]]$.

This tension within the system of linearization must be resolved within the phonology, and as a result, I propose that the reduction of these dependencies is forced by a phonological constraint. The most natural candidate is presented in (80): a faithfulness constraint, drawn from the Correspondence Theory of McCarthy & Prince 1995, 1999, which prevents input elements from having multiple output correspondents.

- (80) *No Pronunciation in Multiple Positions*
 UNIFORMITY: AOV for every element in an input morphosyntactic representation *s* that corresponds to more than one element in a corresponding output phonological representation *p*.

When morphosyntactic representations like (79) are passed off to the phonology, then, it will be the constraint UNIFORMITY that ensures that they will only pronounced in a single position. The task of ensuring that moved elements are generally spelled out in their highest positions, in turn, will fall to the interface constraint in (81). This is a constraint that makes pure reference to the c-command relationships that hold between the various positions of the moved element (as the crucial factor is c-command, not linear order: Nunes 2003; cf. Brody 1995; Bobaljik 2002). Formally speaking, we can state it as a member of the family of constraints on POSITIONAL FAITHFULNESS (Beckman, 1998): that class of constraints that prevent the suppression of elements in structurally prominent positions.

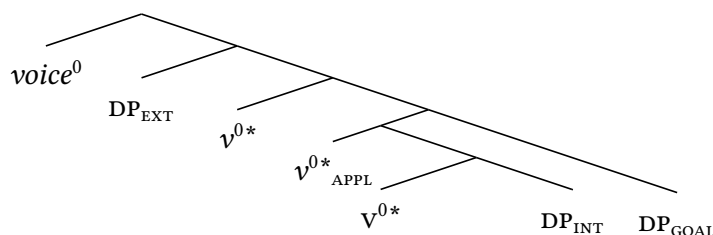
- (81) *Pronunciation in the Highest Position*
 MAX(HIGHEST): AOV for every element that is linearized in multiple positions in an input morphosyntactic representation *s* such that the syntactically highest instance of that element in *s* lacks an output correspondent that is linearized in the same position in a corresponding output representation *p*.

The combined effect of these constraints is to yield the pattern of “highest-copy” pronunciation in (79): within the surface phonology, UNIFORMITY requires moved elements to be realized once and MAX(HIGHEST) prevents non-pronunciation in the highest position. But something more must be said about the cases where moved elements are spelled out in lower positions along their paths: for instance, within the system of covert raising described in Adyghe by Potsdam & Polinsky 2012 and in the ordinary A-syntax of Mandar, where the absolutive DP is usually spelled out in the *vp* even as it raises to SPEC,*voiceP*.

5.3 Spell-Out in the Mandar VoiceP

With all of this in place, we can return to the matter of linearization in the Mandar *voiceP*. I propose that the preference to spell out arguments in their base positions is one that emerges from the idiosyncratic properties of the heads in this space. It is clear that heads can impose idiosyncratic requirements on the linearization of the movement dependencies created by head-movement (Brody, 2000; Arregi & Pietraszko, 2018), and it has also been argued that heads can impose output requirements on the elements they select: Landau 2007, for instance, argues that heads can demand the presence of output correspondents for their specifiers (thus “phonological EPP effects”). As a result, I propose that the Mandar *s*, *o*, and *D* are introduced by heads that impose this type of requirement—canonically forcing these arguments to be spelled out in the *vp*. In the following tree, I mark this requirement with the diacritic *** (roughly equivalent to the [+P] feature of Landau 2007).

(82) *The Mandar VoiceP*



In the phonology, I assume that this requirement is enforced by a constraint of the family SUBCAT, the type of pressure responsible for enforcing item-specific requirements more widely in the phonology (Mascaró, 2007). The operative constraint is given in (83).

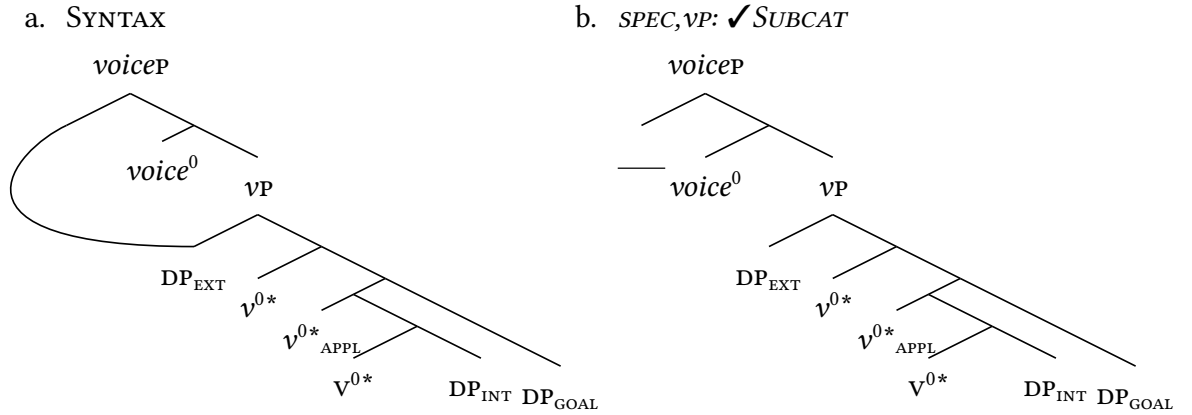
(83) *Head-Specific Requirements*

SUBCAT: AOV for every head x^0 with a specifier *YP* in an input morphosyntactic

representation s such that there is no output correspondent to YP that is linearized in the same position as YP in a corresponding output representation p .

When *voiceP*s of the shape in (82) are sent to the phonology, we can take the surface pattern of spell-out to emerge from the phonology-internal ranking of $SUBCAT > MAX(HIGHEST)$. The following trees illustrate: in a clause where the s raises to $SPEC, voiceP$ in the presence of an antipassive verb (84a), the s must be realized in $SPEC, VP$ to satisfy the output requirements of the head v^0 , rather than being realized in $SPEC, voiceP$ to satisfy the more general requirement to pronounce moved elements in their highest positions (84b).

(84) *Spelling out Arguments in the Mandarin voiceP*

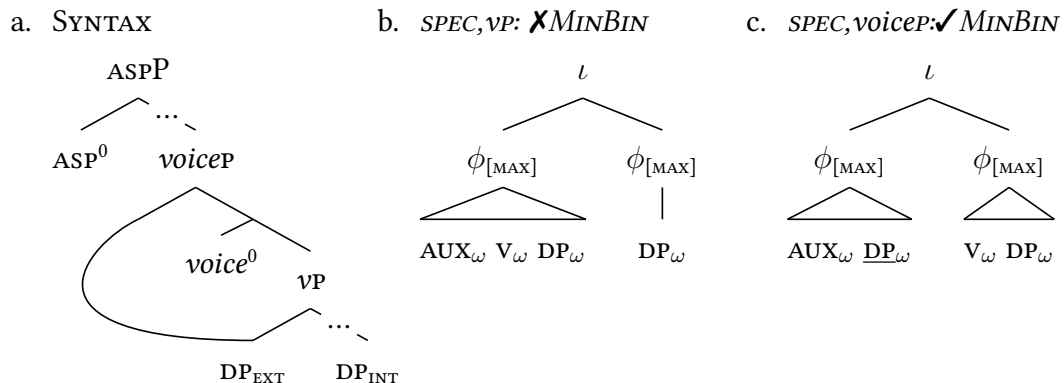


The result is an account that allows us to derive the typical pattern of linearization in the language from the idiosyncratic properties of the functional heads along the clausal spine. In broader perspective, in turn, the same account might then be extended to the range of similar systems that have been proposed to operate in other verb-initial languages of the region. Tagalog, for instance, shares two properties with Mandarin: (i) that language places the absolutive DP in no consistent linear position but (ii) consistently requires it to raise covertly to the highest position in the clause (either $SPEC, TP$ (Murasugi, 1992; Maclachlan, 1995; Maclachlan & Nakamura, 1997) or $SPEC, VP$ (Aldridge, 2004)). From the perspective developed here, it seems natural to imagine that this pattern may also emerge from the interface properties of the heads that make up the Tagalog *voiceP*.

We now return to the system of Absolutive Preposing. In the contexts where spell-out of the absolutive DP in its base position would force a violation of Minimal Binariness, as we have seen, the language requires the absolutive DP to be spelled out in $SPEC, voiceP$. Within the framework developed here, this effect must emerge from the phonology-internal ranking of $SUBCAT$ beneath MINIMAL BINARITY: in other words, the “phonological EPP effects”

that force arguments to be spelled out within the *vp* must be outranked and overruled by the phonological well-formedness constraints that govern the prosodic structures built around different patterns of spell-out. The following trees show this effect: in a clause where the *s* raises to *SPEC,voiceP* (85a), the pressure to realize the *s* in *SPEC,vp* is overruled by Minimal Binarity (85b), which forces the *s* to be realized in *SPEC,voiceP* instead (85c).

(85) *Absolute Preposing: Minimal Binarity > Subcat*



6 Conclusion

With this much in place, we can now advance upon the final network of conclusions that emerge from the results we have secured. The first of these is a fact about Mandar itself. In the clauses that lack auxiliaries, we now have the phonological know-how to set up the prosodic tension that drives Absolute Preposing: by introducing another word into the *s* in a ditransitive *vso* clause, for instance, we can force the *INT* to become a single-word ϕ_{MAX} (86a). When we replicate this move in a clause where the *INT* is absolutive, we then see an interesting repair: the *INT* moves left, but in this context, it falls after the *v* (86b).

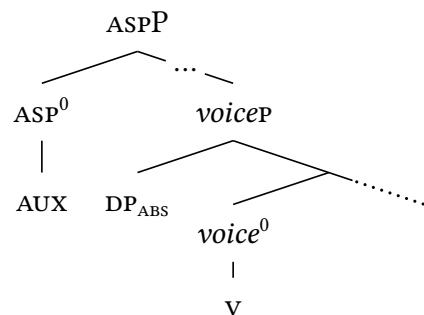
(86) *Absolute Preposing in Verb-Initial Clauses*

- a. $\{\phi_{\text{MAX}}\}$ $\{\phi_{\text{MAX}}\}$ $\{\phi_{\text{MAX}}\}$ $\{\phi_{\text{MAX}}\}$
- | | | | | | | | |
|-----------------------|------|------|-------------------|--------------------|--------------------|----------------------|----------------|
| nasakáŋa ^H | ĩ | ɬo | pósa ^H | maméa ^H | baláo ^H | djónɲɪŋ ^H | ó ^H |
| na-saka-ŋaŋ | i | do | posa | mamea | balao | dionɲɪŋ | o |
| 3ERG-catch-APPL | 3ABS | that | cat | red | mouse | yesterday | there |
- ‘That red cat there caught the mouse for him yesterday.’
- b. $\{\phi_{\text{MAX}}\}$ $\{\phi_{\text{MAX}}\}$ $\{\phi_{\text{MAX}}\}$ $\{\phi_{\text{MAX}}\}$
- | | | | | | | | | |
|---------------------|------|--------------------|------|-------------------|--------------------|---|----------------------|----------------|
| nasáka ^H | i | waláo ^H | do | pósa ^H | maméa ^H | — | djónɲɪŋ ^H | ó ^H |
| na-saka | i | balao | do | posa | mamea | | dionɲɪŋ | o |
| 3ERG-catch | 3ABS | mouse | that | cat | red | | yesterday | there |
- ‘That red cat there caught the mouse yesterday.’

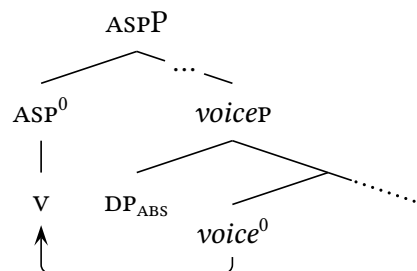
This fact suggests the following: in the clauses that lack auxiliaries, the absolutive DP still raises and can still be spelled out in *SPEC,voiceP*, but that high position follows the verb in this context because in the absence of auxiliaries the Mandarin verb raises past *voice*⁰.

(87) *Verb Raising in Mandar*

a. WITH AUXILIARIES



b. WITHOUT AUXILIARIES



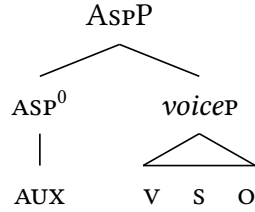
The second conclusion surrounds the particular syntax of the absolutive DP. We have seen through our investigation that Mandarin must belong to that class of languages where the absolutive DP raises to the highest clause-internal A-position, and through our study of Absolutive Preposing, we have come to relatively direct evidence that along the way it must transit through the edge of the *voiceP*. On the understanding that the *voiceP* corresponds to the clause-internal phase, then, we arrive at an important type of confirmation—in Mandarin, at least—that the clause-internal phase head plays a central role in allowing this type of non-local A-movement to occur. This is a position that has been widely adopted in work on other High Absolutive systems, both in Austronesian (Rackowski, 2002; Aldridge, 2004; Nomoto, 2013, 2015; Erlewine & Sommerlot, To Appear) and in Mayan (Coon *et al.*, 2014, 2021; Royer, 2022, 2023), but it is one that has been very rarely substantiated with explicit evidence from linear order—as the best-studied High Absolutive systems show verb-initial word order and provide little linear evidence for the path of absolutive DP.

The third conclusion surrounds the nature of the interface of phonology and syntax. Within the wider literature on that interface, the phonological pressure at the heart of our analysis—MINIMAL BINARITY—is well-attested and well-understood. But many of the fine parses that have emerged through our investigation have a relatively surprising shape, and the most important of these is the parse in (88): that which emerges in four- ω clauses that contain auxiliaries where Absolutive Preposing cannot apply. In strings of the order AVSO, for instance, the syntax in (88a) is never mapped to the isomorphic prosody in (88b), where the *voiceP* is parsed into a ϕ . Instead, this string receives the mismatched parse in

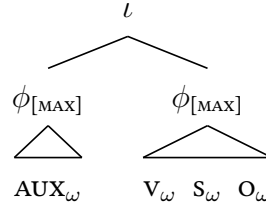
(88c): the verb is parsed together with the AUX and the s, prosodically stranding the o.

(88) *The Parse of AVSO*

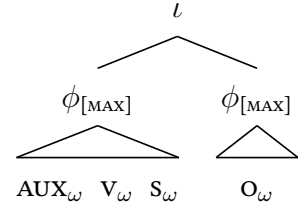
a. *Syntax*



b. *Expected Parse*



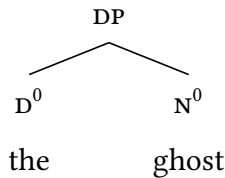
c. *Attested Parse*



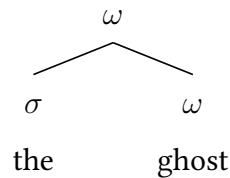
The emergence of parse (88c) suggests that the syntax-prosody mapping must be guided by two types of pressure beyond the requirements of isomorphism and eurhythm. The first is the pressure, noted by Kalivoda 2018, to parse verbs into ϕ s with as many arguments as possible when weight constraints prevent the construction of ϕ s around complete xPs like the *voiceP*. The second is a parallel requirement to parse lexical heads into ϕ s with the functional elements in their extended projections, like the AUXes above. This second pressure is likely the source of a second well-known but poorly understood pattern at the interface: heads are often parsed into words with adjacent functional elements in their extended projections (Booij, 1996; Peperkamp, 1997; Itô & Mester, 2009). Thus even without head movement, FNC-LEX sequences like (89a) receive parses like (89b).

(89) *The Parse of Fnc-Lex Sequences*

a. *Starting Syntax*



b. *Attested Parse*

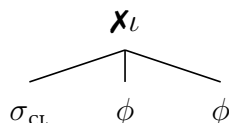


Our final conclusion surrounds the nature of the interaction between prosodic phrasing and linearization. We have situated our analysis of Mandar within a network of cases in which prosodic phrasing seems to determine the positions in which moved elements are spelled out, drawing a parallel with the analysis of second-position clitics in Serbo-Croatian in Franks & King 2000 and Bošković 2001. But there is a difference between the facts of Serbo-Croatian and the patterns studied here: in Mandar, but not Serbo-Croatian, the elements whose positions are determined by prosodic constraints are prosodically independent and syntactically phrasal constituents. This fact is important because there is

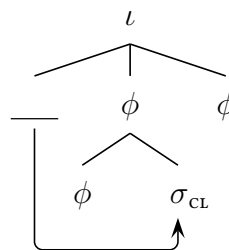
a way to capture the linear positions of second-position clitics in Serbo-Croatian with no reference to the architectural model that we have developed above. This is by positing a type of movement in the phonology, along the lines of [Halpern 1995](#), which ignores the syntax and simply postposes these elements from the left edge of the intonational phrase.

(90) *Phonological Movement of Second-Position Elements*

a. IMPOSSIBLE LINEARIZATION



b. PHONOLOGICAL REORDERING



The case for this type of analysis has been steadily strengthened by a growing body of evidence to suggest that the phonology can indeed force the movement of prosodically deficient elements, either in response to item-specific requirements ([Chung, 2003](#); [Aissen, 2017](#)) or general output constraints ([Bennett et al. 2016](#); [Kusmer 2020](#); and in the second-position systems of Serbo-Croatian, [Diesing & Zec 2011, 2017](#); Tagalog, [Kaufman 2010](#); and Mandarin, [Brodskiy 2021b](#)). But there is good reason to avoid a fully phonological analysis of Absolute Preposing (e.g., along the lines of ([Agbayani & Golston, 2010, 2016](#); [Agbayani et al., 2015](#)), as the alternations in that system track the fact of syntax in a precise and exacting way. To the extent that our analysis is successful, then, it provides independent evidence that prosodic phrasing must be able to condition the positions in which xps are spelled out along their paths of movement. If movement dependencies can indeed be linearized in a way that is sensitive to prosodic phrasing, then, we arrive at an architecturally important result: the relevant type of linearization must operate at a stage that has access to this information. The balance of evidence in the phonology, in turn, suggests that there is only one derivational moment that fits the bill: the global and parallel phonological evaluation where prosodic structure is built.

Looking beyond the internal architecture of the grammar, at last, our results deliver another line of evidence to support the Minimalist elimination of the *trace*. Within the derivational frameworks that eschew that step, it has been roundly observed, there should be no way for phonological factors to determine the positions in which moved elements are spelled out. The discovery of another case of this type, then, provides a novel argument that this theoretical step is empirically correct.

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